Bybit Options-Income Quant-Coach

Design & Knowledge Compendium

Concepts, models, architecture, and lessons behind a private options-income decision-support system.
Ideas only — no source code.

Compiled 2026

00 — Overview

The problem being solved

Selling options premium on crypto is attractive (high implied volatility, 24/7 markets, weekly and daily expiries) but operationally punishing for a solo trader:

1. **The data is scattered.** Implied-vol surface, skew, term structure, funding, realized vol, open-interest/gamma positioning, and your own live greeks live in five different places. Deciding whether vol is *rich* right now means assembling all of it by hand.
2. **The exchange hides the true cost.** Bybit's option *mark* price is theoretical; the price you can actually transact is the bid/ask. Fees (trading + delivery) quietly eat a double-digit percentage of gross P&L. A position that looks green on mark can be a loss to close.
3. **Discipline decays.** Without a memory of what has actually worked *for you*, every decision restarts from intuition. The edge in premium selling is small and statistical; it only shows up if you measure your own record honestly.
4. **Naked short premium is unforgiving.** The book this was built for runs fully naked

short strangles on BTC — an uncapped-tail posture. That demands a risk lens that never lets a "looks fine on mark" reading hide a tail that is quietly growing.

The system exists to collapse all of that into a handful of screens that answer concrete questions: *Is vol cheap or rich? What structure fits this regime? Which of my legs needs attention? Can I actually get out of this position, and at what cost? Where is my real edge?*

Product philosophy

- **Coach, not autopilot.** The software's job is synthesis, judgment support, and memory. The human keeps the executing hand. This is a deliberate safety and skill-building choice, not a technical limitation.
- **Transparency over black boxes.** Every score is a sum of visible components, each with a reason string that carries its own point contribution. The trader can always see *why* a strategy ranked where it did. No fitted ML weights that can overfit or can't be audited.
- **Honesty about sample size.** Any statistical claim (percentile ranks, personal win-rates, seasonality) is withheld or explicitly hedged until enough data exists to support it. "Only N days tracked so far" is a first-class output.
- **First-party data where it matters.** Rather than scraping third-party visualizers, the system computes its own IV surface from the exchange and accrues its own history. External venues (Deribit) are used only as a labeled *reference*, never as a hidden signal source.
- **The flywheel.** Every trade entered snapshots the full market context at entry. That record cannot be backfilled, so capture starts on day one — and the longer the system runs, the more its calibration is worth.

The non-negotiable constraints (guardrails)

These held for the entire life of the project and shaped every feature:

1. **No order automation.** The read-only API surface is used deliberately: positions and wallet balance only. No trade, transfer, or withdrawal endpoints are ever touched.
2. **Not personalized financial advice.** Output is framed as data synthesis plus the trader's own rules. A disclaimer is always present.
3. **Per-account isolation.** The tool supports multiple sub-accounts with separate data and separate keys; no calculation ever leaks one account's data into another.
4. **Never emit invalid numbers.** `inf`/`nan` must never reach the JSON layer (they break the browser's parser). Undefined ratios are `null`, not infinity.
5. **Clearly flag estimated or missing data.** Anything unverified, placeholder, or soft-failed is labeled as such in the surface that shows it.

Technology at a glance

- **Backend:** Python, an async web framework, an async HTTP client. Pure-function core modules (all the math) separated from the I/O layer so everything is unit-testable without a network.
- **Frontend:** a single self-contained HTML file, vanilla JavaScript, no build step. Tabs for each surface; a multi-account switcher.
- **Storage:** flat JSON files on disk (per-account portfolios and closed trades; global market-history stores). No database.
- **Data sources:** Bybit public V5 (instruments, tickers, candles, linear/perp index & funding), Bybit private V5 read-only (positions, wallet balance), Deribit public v2 (DVOL index + option book summary) as an external reference.

01 — Design

This document is the conceptual core: the architecture and the full library of quant models the system is built on. Formulas are given because they *are* the ideas; no source code is included.

> *Every formula referenced here is stated in full — with its provenance tag and a numbered citation — in 07 — Math & references. Where this document says "Yang-Zhang" or "HAR-inspired" or "the delivery-fee cap", that document says exactly what the expression is and where it comes from.*

1. Architecture

1.1 Two layers, one machine

- **A pure-function core.** Every piece of math — fees, margin, volatility estimators, payoff geometry, scoring, calibration, backtests — lives in modules that take plain numbers and return plain numbers. No network, no disk, no global state. This is what makes the whole system unit-testable against synthetic fixtures with zero I/O.
- **A thin I/O shell.** A small async web layer fetches exchange data, reads/writes the JSON stores, and calls into the pure core. Endpoints are deliberately dumb: fetch, compute, serialize.

The discipline of "math is pure, I/O is a shell" is the single most important structural decision. It is why every model below could be verified in isolation.

1.2 Frontend

- A **single self-contained HTML file** with vanilla JavaScript and no build step. Chosen for zero-friction deployment and total transparency — the entire UI is one file you can read top to bottom.
- **Tabbed surfaces**, each mapping to one decision context (screener, intelligence, portfolio, coach, calculator, etc.).
- **Event delegation** on stable container elements rather than per-row handlers, so live re-renders never steal input focus mid-typing. Network writes are **debounced**.

1.3 Per-account isolation

- Multiple sub-accounts, each with its own data directory and its own API credentials.
- **The race-safety rule:** every request handler captures the active account's directory path into a local variable *before its first `await`*. Because an account switch can land between awaits, reading the "current account" lazily would let one request read account A's positions and write them under account B. Snapshotting up front closes that race. This one rule is applied to every account-scoped endpoint without exception.

1.4 Storage

- **Flat JSON files.** Per-account: the open portfolio and the closed-trade ledger. Global (market data, not account-scoped): the daily regime history and the intraday IV history.
- No database — the data volumes are small, the access patterns are simple, and flat files are trivially inspectable and backup-able.
- **Global vs per-account is a deliberate split.** Market history is shared across all accounts because the market is the same for everyone; only positions and outcomes are private.

1.5 The background snapshot loop (the only "automation")

- A single background task wakes periodically and, for each underlying, captures a snapshot of the market context if one is due. It feeds **two stores at two cadences off one computation**: the daily regime store (once per UTC calendar day) and the intraday IV store (once per session window).
- **Idempotent capture:** writing "today's" snapshot replaces the existing row for that key rather than appending, so restarts and multiple fires never corrupt history.
- **Gaps are tolerated, never backfilled.** If the server was offline, that day is simply missing; the system never fabricates history.
- This is read-only capture. It is the *only* automated behavior in the entire system, and it touches nothing but its own history files.

2. Data sources

2.1 Bybit public (V5)

Option instruments and tickers (mark, mark-IV, bid/ask, greeks), daily candles, and linear/perpetual index price and funding history. The option chain is assembled by merging the instruments list with the tickers list into a unified contract object, filtered by days-to-expiry, and briefly cached.

2.2 Bybit private (V5), read-only

Only two endpoints: current option positions and unified wallet balance. HMAC-signed requests. **No trade, transfer, or withdrawal endpoint is ever called** — this is enforced by simply never implementing them in the client.

2.3 Deribit public (v2) — external reference only

Two unauthenticated calls: the DVOL volatility index and the option book summary. Used to answer "are we rich or cheap versus the deepest-liquidity venue?" — always labeled as an external reference, never fed silently into any signal.

2.4 The index-price gotcha

Bybit option tickers carry an `underlyingPrice` field that is **unreliable**. Spot is taken instead from the **linear perpetual's** `indexPrice`. Getting this wrong poisons every downstream calculation (OTM amount, margin, moneyness, greeks), so it is centralized.

3. The quant concept library

3.1 Options fee model

Realized P&L is computed **net of exchange fees**, because on a premium-selling book fees are not a rounding error — on the reference book they ran roughly **13% of gross P&L**.

- **Trading fee per fill** = `min(rate × index, 7% × option_price) × size`
  - taker rate ≈ 0.03%, maker ≈ 0.02%.
  - The **7%-of-premium cap** is the subtle part: for far-OTM cheap legs the percentage-of-premium term is smaller than the percentage-of-index term, so the cap binds. Ignoring it overstates the fee on exactly the legs a strangle-seller trades most.
- **Delivery (settlement) fee** = `min(deliveryRate × index, 12.5% × intrinsic) × size`, charged **only on exercise** (in-the-money at expiry); an out-of-the-money expiry is free.
  - `deliveryRate` ≈ 0.015% for BTC/ETH, ≈ 0.02% for alt underlyings.
  - **Exception:** *Daily* options incur **no** delivery fee. This single exception matters for breakeven math (below) and was re-verified against the exchange's own help page.
- A round-trip close pays trading fees on **both** the open and the close fill. Modeling only one side understates cost by half.

3.2 Initial-margin model (short options)

Per unit, for a short option:

```
IM = max( MaxIMFactor × index − OTM_amount , MinIMFactor × index ) + max( avg_price , mark )
```

then multiplied by size. Where:

- `OTM_amount(call) = max(0, strike − index)`, `OTM_amount(put) = max(0, index − strike)`.
- `MaxIMFactor`, `MinIMFactor` ≈ (0.10, 0.05) for BTC/ETH (verified against the exchange's worked example). Alt-coin factors are higher and live on a dynamically-rendered exchange page; where they can't be scraped, the BTC/ETH values are used as a **clearly-flagged placeholder** and the risk summary emits an "estimated" warning.
- A long option's margin is simply its premium.

Why hand-roll this: the unified account reports empty per-position IM/MM for options; the real figures only come from the account-level wallet balance. To attribute margin *per leg* (so the trader can see which legs to trim), the formula must be reconstructed. An early version used a wrong delta-proxy and was ~2x off on deep-OTM legs — the corrected OTM-scan formula above matches the exchange to the cent.

3.3 Contract-size convention

- Quantity is always stored in **underlying units** (e.g. BTC), not contract count.
- `1 contract = contract_size` underlying units: 0.01 for BTC/ETH, **1.0 for SOL** (a real

gotcha — SOL's minimum order quantity is 1, not 0.01).

- Because qty is in underlying units, the margin formula is **size-agnostic** — it needs no knowledge of contract size. Only *contract count* and per-contract sizing need it. Keeping this straight prevents a whole class of 100x errors.

3.4 Volatility models

- **Realized-vol estimators** from OHLC candles: Parkinson (high-low range), Garman-Klass (adds open/close), and **Yang-Zhang** (adds overnight gaps; the primary estimator, most efficient for gappy 24/7 crypto).
- **RV forecast, HAR-inspired:** a fixed-weight blend of short/medium/long realized vol (weights 0.3 / 0.4 / 0.3 over 5 / 20 / 60 days of Yang-Zhang). The weights are **fixed and visible, not fitted** — deliberately un-optimizable so they cannot overfit.
- **Variance/volatility risk premium (VRP)** = `IV - forecast_RV`, reported both in vol points and as a ratio. This is the core "is selling premium worth it right now" gate.
- **Expected move** per expiry from the ATM straddle price, compared against the realized distribution of absolute moves over the same horizon — answers "is the market pricing more movement than tends to happen?"
- **Trend filter:** fast/slow moving-average relationship with a **dead-band** (~0.5%) plus a short lookback return. The dead-band is essential: a zero-tolerance MA comparison flips label on noise. (A test caught exactly this.)
- **Funding summary:** perpetual funding annualized from recent 8-hour rates — a positioning and carry signal.

3.5 ATM IV versus surface-average IV (a load-bearing distinction)

Averaging implied vol over *every* contract (all expiries, the whole smile) produces a number dominated by the expensive wings — it can read ~41% when the true at-the-money vol is ~33%. That surface-average is **not comparable** to any single-point reference like DVOL or another venue's ATM.

- The correct "IV" for cross-venue comparison and for headline display is **ATM IV**: the near-the-money implied vol at a reference expiry (the soonest with a few days to go, to skip the noisy 0-DTE smile).
- Mislabeling the surface-average as "ATM IV" and comparing it to an external ATM reference manufactures a phantom "we're 9 points rich" signal. **Always compare ATM-to-ATM.**

3.6 Skew (25-delta) and its sign convention

- **25-delta skew = put IV − call IV** at equal delta magnitude. **Positive = puts richer** (the normal fearful state).
- The sign convention must be applied *identically* everywhere it is consumed (labeling, digest signals, scoring). A sign bug that treated negative as put-rich labeled every fearful, put-rich market as "call skew" and scored it backwards — live and invisible until

audited. The lesson: define the sign once, document it at every consumption site.

- When computing skew from a reference venue, the delta is reconstructed via Black-Scholes (all inputs are present) rather than trusted from a coarse strike grid, and the value is **withheld** when the nearest available strike isn't genuinely near 25-delta — a coarse grid must not be allowed to lie.

3.7 IV rank and term structure

- **IV rank** = where current IV sits within its own recent history, as a true percentile. Early versions used realized vol as a proxy denominator; the system now accrues real IV history (below) so the rank becomes a genuine self-referential percentile over time.
- **Term structure** = the shape of ATM IV across expiries (backwardation vs contango), which flips the attractiveness of calendar structures.

3.8 Executable pricing — the mark is not a price

The most practically important insight in the whole system. It applies at **both ends** of a trade, and the system got the exit right long before it got the entry right.

3.8a Entry: what a structure actually fills at

As a taker you never get mid. You get the side of the book that is worse for you:

- **SELL a leg → you receive the BID.**
- **BUY a leg → you pay the ASK.**

So a short strangle collects `bid_put + bid_call`, not `mid_put + mid_call`; a bull put spread collects `bid_short - ask_long`. On the wide, deep-OTM books this account trades the difference is not a rounding detail — **for a put quoted 5 / 25 the mid is 15 and the bid is 5, so two thirds of the "credit" a mid-based screener reports is spread handed to the market maker on entry.** Every fill also pays a trading fee, charged on *every* leg — a four-leg iron condor pays it four times on the way in.

The screener therefore reports **three numbers side by side**, so the gap cannot be missed:

	meaning
`net_credit_mid`	the optimistic number a mid-based screener shows
`net_credit_exec`	after crossing the spread: bid on shorts, ask on longs
`net_credit`	after crossing **and** after entry fees — **what you keep**

`net_credit` is the headline **deliberately**: it is the only one of the three that describes money, and every downstream figure (breakeven, max loss, annualized yield, probability of profit) is derived from it, so a candidate's numbers reconcile with each other. The identity

$$textttnet\_credit\_mid - textttnet\_credit = texttspread\_cost + texttentry\_fees$$

always holds, which is what makes the decomposition auditable rather than merely pessimistic. **Edge retention** = `net_credit / net_credit_mid` is the fraction of the paper credit that survives contact with the order book: a structure retaining 90% is a real trade, one retaining 20% is mostly a gift to the market maker. (Formal statement: 07 §8.6.)

Two failure modes are named rather than smoothed over:

- `credit_vanishes` — mid calls it a credit trade and the real book calls it a debit.
- `quote_estimated` — the venue publishes no quote on the side this fill needs (normal on short-DTE deep-OTM contracts), so the price fell back to mid/mark. The number is then a *model* price, not a fillable one, and the caller is told so rather than shown a plausible fiction.

Exit costs are deliberately excluded from entry pricing. What it costs to get out depends on when and why you leave — expiring worthless is free, closing early crosses the spread again and pays another fee — and that belongs to the roll analyzer (§3.20) and the P&L calculator, not to a screener ranking entries.

One source of truth. The screener and the roll analyzer originally carried separate copies of the same three helpers, which is exactly how a fix to one silently misses the other. They now share a single pricing module; the identity above is what a test asserts.

3.8b Exit: the executable close quote and Capture%

- Bybit's option **mark is theoretical (a model mid)**. It is *not* a price you can trade.
- To **close** a position you cross the spread: a short buys back at the **ask**, a long sells at the **bid**. That executable quote is what determines whether you can actually exit and at what cost.
- On deep-OTM legs the mark can be near zero (e.g. 0.02) while the only ask is a wide, one-sided quote (e.g. 5). "Slippage versus mark" therefore explodes to thousands of percent and is meaningless. Two corrected readings replace it:
  - **Crossing cost** measured as the **half-spread versus the bid-ask mid**, which never explodes; and a **one-sided flag** when only the side you must hit is quoted (no opposite quote → fill price is uncertain, use a limit).
  - **Capture%** = the fraction of the premium you actually keep if you close *now at the executable quote*: `(entry - buyback_ask) / entry` for a short (mirrored for a long). This is the honest cousin of the mark-based "profit%"; the gap between the two columns is the *cost of reality* (stale marks plus the spread). A leg can show a healthy mark-profit yet a negative capture — meaning closing it actually realizes a loss.
- **Color follows economics, not liquidity.** The close-price color tracks Capture% (green only when you keep a majority of the credit), while spread width lives as a subordinate tag. An earlier version colored by spread width and painted losing-but-tight legs green — a genuine confusion that the redesign fixed.

3.9 Fee-adjusted breakeven (why it needs a root-finder)

The at-expiry underlying price where a leg nets exactly zero **after** the open trading fee

and the delivery fee is not a closed-form expression. Because the delivery fee is `min(rate × index, 12.5% × intrinsic)`, *which term binds depends on the premium size* — cheap OTM legs hit the intrinsic cap, large legs hit the index term. So the breakeven is solved by **bisection against the exact fee model**, not a formula.

- A short put's fee-adjusted breakeven shifts **up** toward the strike (less room); a short call's shifts **down**; longs need a bigger move to overcome the fee drag.
- For *daily* options the delivery term is dropped entirely, so the breakeven collapses to a clean `strike − (entry − open_fee)`.

3.10 Payoff engine (piecewise-linear)

A generic engine computes **exact breakevens, max profit, max loss, and reward:risk for any combination of legs** by evaluating the piecewise-linear expiry payoff — the kinks are at the strikes, so the whole curve is determined by its value at and between strikes. Verified against the closed-form payoffs of standard structures. Cross-expiry structures (calendars) are excluded because their payoff isn't piecewise-linear at a single expiry.

3.11 Gamma positioning: GEX, max pain, zero-gamma flip

- **Max pain** = the strike minimizing total option-holder value (where the most open interest expires worthless) — a soft magnet into expiry.
- **Gamma exposure (GEX) profile** across strikes classifies each of your strikes as sitting in a dealer **long-gamma** zone (mean-reverting, pin-safe) or **short-gamma** zone (trend-accelerating). Heavy put open interest flags short-gamma; heavy call OI flags long-gamma.
- **Zero-gamma flip** = the spot level where the cumulative net-GEX profile crosses zero — the boundary between the stabilizing and accelerating regimes. Used as a summary field, a digest signal, and a scoring input.

3.12 Regime detection and memory

- Each underlying's regime is summarized from IV rank, VRP, trend, skew, put/call ratio, funding, and term structure, and condensed into a one-line **named regime banner** (e.g. "High-IV, range-bound, contango, premium-rich"), tinted by VRP.
- **Historical context via percentile rank:** today's reading is ranked against the book's own recent history (e.g. last 90 days), and a 5-day movement arrow is attached. Both are **withheld or hedged below a minimum sample** — "only N days tracked so far" instead of a percentile the data can't support.

3.13 Strategy playbook — transparent additive scoring

- Every candidate strategy starts at a **base score of 50**. Each signal component (VRP, IV rank, trend, skew, put/call ratio, funding, term structure, expected-move richness, gamma regime) adds or subtracts **visible points**, and each point contribution is carried inline in a human-readable **reason string**. Scores are clamped to a sane band.

- **Defined risk is structurally favored:** naked/undefined-risk structures carry a small fixed penalty plus a tie-break, so at equal signal edge a defined-risk structure outranks a naked one.
- The top pick is emitted, or **"Wait"** when nothing clears an action threshold — but the full ranked list is always attached so the UI can show *what was considered and why*.
- Because scoring is pure addition of labeled components, it is completely auditable. There is no fitted model and no hidden weight.

3.14 Strategy library (the structures scored)

A library of ~11 generators, each a pure function over a chain: cash-secured put, covered call, bull-put and bear-call credit spreads, iron condor, short strangle, **jade lizard** (short put + short call spread, emitted only when the credit ≥ the call-spread width so upside risk is structurally zero), **broken-wing butterfly** (credit put fly harvesting put skew), **put ratio spread**, **calendar** (the only long-vega/debit structure — it *inverts* the VRP/IV-rank/term-structure scoring because it wants cheap vol and contango, so low-IV regimes surface a calendar instead of "Wait"), and defensive structures: **put ladder** (short put + two lower longs, credit-only, so a crash turns *profitable*), **iron butterfly** (ATM pin play), and **seagull** (zero-cost-or-credit directional participation). Each is liquidity-filtered and payoff-annotated.

3.15 Personal-edge calibration (the learning loop)

The system leans its advice toward what has actually worked for the trader — bounded, transparent, and sample-gated:

- Closed trades are **attributed by the short leg**: a short put credits the put-selling families (CSP, bull-put, jade, broken-wing, put-ratio); a short call credits the call-selling families. Range structures (condor, strangle, calendar) are honestly **un-attributable** from leg data and get no nudge.
- The nudge is computed from **win-rate and expectancy**, and is **regime-aware**: it narrows to trades entered within a ±20 IV-rank band of the current regime when enough exist, else uses all history.
- It is **expectancy-gated** (a high win-rate can't rescue negative expectancy) and **hard-capped** (roughly ±6 points) so market signals still dominate. A minimum trade count gates it on entirely.
- When applied, each nudge adds its own reason line carrying the actual record (e.g. "your put-selling in similar IV: 8-2, +142 avg → personalized +6").
- **The flywheel that feeds it:** every entered leg snapshots the full market context at entry (IV rank, IV/RV, skew, term structure, put/call ratio, VRP, funding, trend, regime

label, spot). This snapshot **cannot be backfilled**, so it is captured from day one and becomes more valuable the longer the system runs. Stored *market* history stays pure — the calibration feedback is applied only at the advice layer, never written back into the recorded regime data.

3.16 Carry monitor

Compares three sources of yield to decide where the income is: **dated-futures basis** (annualized, cash-and-carry), **perpetual funding**, and the **VRP** (option selling). A rule-based verdict picks carry / vol / both / neither against explicit bars (e.g. basis ≥ 5%/yr, VRP ratio ≥ 1.05). A venue gotcha shaped this: the exchange **ignores the base-coin filter on linear tickers**, returning every contract, so results must be filtered by symbol prefix.

3.17 Backtesting methodology

- **Walk-forward with no lookahead**, enforced by a strict slicing convention: "the world as of day *t*" is only the data available up to *t*. Forward outcomes are measured on strictly later bars. This rule is the difference between an honest backtest and an accidental oracle.
- **Realized-vol studies:** HAR-blend forecast vs a naive trailing estimator (error/bias/beat-rate); trend-label to forward outcomes (the "falling-knife" guard's report card); and sigma-based strike touch/finish probabilities versus one-sided normal theory.
- **IV studies**, unblocked by a reference venue's multi-year DVOL history: the structural VRP edge (implied vs tenor-matched realized-forward) and the predictiveness of IV rank (does "sell when IV rank is high" pay?).
- **Headline empirical findings** (the reference book, BTC):
  - The **ATM vol-risk premium is razor-thin** (implied ≈ realized-forward over ~30 days) — so this book's edge is **not** ATM richness but the **OTM skew/tail it sells** (naked strangles) plus theta/gamma management.
  - **IV-rank timing does add** a couple of vol points (sell-high-IV-rank validated); the mid tier is noisy from overlapping windows and is reported as such.
  - Crypto **tails are ~2.7x Gaussian** at monthly horizons (a 2-sigma/30-day put finishes in-the-money ~6% of the time vs ~2% under normal theory) — an argument for shorter DTE or wider strikes on long-dated naked puts.

3.18 Risk sizing

- **Vol-targeted sizing:** scale position size by `min(1, target_vol / realized_vol)` with a floor, so exposure shrinks when the market gets loud. The realized-vol feed is the latest regime snapshot (disk read, no network).
- **Capital and margin limits** convert a capital figure and a max-utilization setting into a contracts-allowed cap; a result of `0` is a real "your limits don't allow this at current

capital" signal, distinct from `null` (capital not configured).

3.19 Book-sync / reconciliation

On a cadence, the local book is diffed against the exchange's live positions to detect three kinds of drift: **new** legs present on the exchange but not tracked, **size-changed** legs (a partial close or add), and **closed-early** legs gone from the exchange before expiry. The trader is prompted to import/update or to record the close. The diff runs on every book reload so any action clears or refreshes it immediately, rather than lingering until the next timer tick.

3.20 Roll economics — price it, score it, then check your own record

Rolling is the single most consequential recurring decision on a short-premium book, and the one most easily rationalized. The system attacks it in three layers, each answering a question the previous one cannot.

Layer 1 — what does this roll actually pay?

For a short leg that is tested or near expiry, the analyzer tables the concrete choices from the live chain: **roll out** (same strike, later expiry) and **roll out & adjust** (strike moved further OTM toward a defensive delta target, later expiry).

A roll **crosses the spread twice** — buy the old leg back at the ask, sell the new one at the bid — and pays a trading fee on **both** fills. Mid-based roll numbers therefore overstate the payment systematically; on a wide deep-OTM book the entire apparent "credit" can be slippage. The same three-number decomposition as §3.8a applies, with `mid_flatters` flagging the case where mid says credit and reality says debit.

Per-day figures, not the raw credit, are the comparison. The variance risk premium's term structure slopes steeply downward, so rolling further out for a bigger *absolute* credit often buys **less premium per unit of time-risk**. Candidates at different expiries are compared on credit-per-day and yield-per-day (credit per day as a percentage of collateral), never on the headline number.

Layer 2 — is it a *good* roll, not merely a paying one?

A roll can pay a fat credit and still be a bad idea: it may be selling cheap vol, moving the strike inside one expected move, or doubling the margin to buy a few days. Five independent axes are scored 0–100 and weight-averaged, and **the full breakdown is always returned**, so the trader can disagree with any single axis instead of trusting an opaque rank:

Axis	Weight	What it asks
carry	0.25	credit per day on collateral, annualized — are you paid enough per day?
vrp	0.25	the **new** contract's IV vs forecast RV — is what you're selling actually rich?
safety	0.25	distance from spot to the new strike **in expected moves**

Axis	Weight	What it asks
gamma	0.15	improvement in $|\theta|/|\Gamma|$ — the mechanical reason to roll out of expiry week
capital	0.10	margin change — a credit that doubles initial margin is not free

Three design points carry the section:

- **Safety is measured in expected moves ($\sigma\sqrt{T}$), not in %OTM.** Raw distance is not comparable across tenors. A consequence worth stating because it surprises people: a **same-strike roll scores *worse* on safety as the tenor grows** — correct, because it buys *time*, not *safety*, while the move budget grows with $\sqrt{T}$.
- **The VRP axis prices the contract you are opening, not the one you are closing.** "Rolling into cheap vol to fix an old problem" is the classic losing roll, and it is only visible if the new leg's IV is checked against forecast RV.
- **Gates are reported separately and never folded silently into the score.** A debit roll, or one selling below forecast RV, caps the verdict at *questionable* no matter how the weighted average lands — so a high score with an open gate still reads as a warning rather than as approval.

Weights renormalize over whichever axes are computable, so a missing gamma feed degrades the score's resolution instead of silently zeroing an axis.

Layer 3 — across your history, do rolls create value or postpone losses?

Layers 1 and 2 are about one decision in front of you. Layer 3 is the only one that can answer *"does rolling work for me"*. Legs are linked by a **roll chain**: the original position, then each replacement created by rolling it.

- **Martingale detection.** Rolling a tested position repeatedly — especially while increasing size — is the classic way an income book with an uncapped tail dies. Each roll individually looks like "collect more credit"; **the chain is what reveals that fresh risk has been financing a losing position.** Warnings fire on structure, not prediction: rolled ≥3×; size grew across the chain ("adding size to defend a position is doubling down, not managing risk"); the chain's realized total is negative; every roll in the chain was made while losing.
- **Personal evidence.** Completed chains are split by **the state at roll time** — rolled while in profit (redeploying a winner) versus rolled while tested (defending a loser) — and both are compared against positions never rolled. Two rules keep that comparison honest: only **completed** chains are scored (an open chain's P&L is not decided yet, and counting it flatters whichever way the position happens to be sitting), and a chain containing **both** kinds of roll counts as *defensive* — the defensive roll is the risk-relevant event, and treating a mixed chain as clean would bias the comparison in rolling's favour.
- The roll context is **captured on the new leg at roll time**, never reconstructed

afterwards, because "was I in profit when I rolled?" cannot be recovered from a closed ledger.

- Below the sample gates the report shows **counts and withholds the verdict**.

3.21 Theta efficiency — is the carry actually being paid for?

The portfolio table shows theta per day, but that number alone says nothing. **Theta is not income: it is the rent collected for carrying gamma.** Over a day a short option earns roughly $|\Theta| - ½\Gamma(\Delta S)^2$, so the move that exactly consumes a day's theta is $\Delta S^* = \sqrt{(2|\Theta|/\Gamma)}$.

The trap this module exists to avoid. It is tempting to compare ΔS^* against the IV-implied daily move. That comparison is **circular**: Black–Scholes fixes $\Theta = -½\sigma^2 S^2 \Gamma$, so substituting collapses ΔS^* into the implied move itself. Verified on the live chain, the ratio sits at a flat **≈0.96 across every strike and every tenor** — it carries no information at all. (Derivation: 07 §9.2.)

The comparison that does carry information is against *realized* movement. That is the variance risk premium measured on *this specific leg with its own greeks*, rather than as a surface average — and it is the honest answer to "is this theta worth it". Realized above breakeven means the position **loses in expectation even while theta prints positive every day**, which is precisely the failure mode a theta-focused dashboard would otherwise hide.

Two by-products fall out of the same numbers:

- **A stale-greeks detector.** Because the ratio in the trap above *must* be ≈0.96, any leg wandering far from it has stale or inconsistent greeks, and its efficiency read is flagged untrustworthy. A degenerate identity turned into a data-quality check.
- **Fragility to a vol spike**, as the number of days of theta one vol point costs ($|v|\ q\ /\ \Theta$). A leg needing many days to earn back a one-point IV rise is fragile regardless of how good its carry looks.

Alongside these: theta per unit of margin (comparable across legs of different sizes) and days-to-collect the remaining premium. When no realized-movement baseline is available the verdict is `unknown` — the system does not substitute implied, because that is exactly the circular comparison above.

The units trap, stated because it is invisible. On an enriched leg theta is *position* theta (already multiplied by quantity) while gamma and vega are *per-unit*. Mixing the two scales leaves a stray quantity inside the square root and produces a plausible, wrong breakeven. Everything converts to per-unit first.

3.22 The monthly-income layer — earn, support, and don't chase

Everything else in the system measures a **trade**: expectancy, win rate, profit factor, hold time. *"Can I live off this?"* is a different question — it is about **time and capital, not about trades** — and none of the per-trade statistics answer it. A book can have an excellent expectancy per trade and still be unusable as an income stream if the month-to-month path is violent enough.

Three layers, each answering the question the previous one exposes.

A — What did the months actually do?

Realized P&L bucketed by the calendar month a trade **closed** in, net of fees, plus:

- **Return on the margin that was tied up to earn it.** 500 on 5,000 of margin and 500 on 50,000 are not the same business. Margin at entry is reconstructed with the same formula the risk tab uses, which needs the index at entry — so trades opened before entry-context capture existed are **reported as uncovered rather than guessed at**, with the coverage percentage stated.
- **Consistency, expressed as the number that actually decides it.** Not the average: **how many average winning months the worst month erased.** For a negatively-skewed short-vol book that ratio is what determines whether the income is spendable.
- **Drawdown on the cumulative curve**, with whether and how quickly it recovered.

Two honesty rules are load-bearing:

- **The current month is incomplete and is excluded from every statistic** — including it flatters or damns the record at random. It is still *shown*, flagged as partial.
- **This is a realized-P&L curve, not account equity.** Open positions are not marked. A short-vol book can show a beautifully smooth realized curve while carrying a large unrealized loss it has not taken yet — which is precisely the failure mode a "monthly income" framing encourages. Every report says so.

And the gate: monthly statistics need **months, not trades**. A hundred trades inside three months is still three data points, so below a minimum of complete months every derived statistic is withheld and the reason is printed.

B — How much can come out before the program stops surviving?

Fitting a distribution to a dozen observations of a negatively-skewed, fat-tailed return stream produces a clean number worth nothing. So the system **bootstraps from the months actually observed** — resampling with replacement, compounding equity proportionally, taking the withdrawal in cash afterwards, seeded so the answer does not move between reloads.

That buys honesty at a price which is restated every time it is shown:

> ***A bootstrap cannot produce a month worse than the worst one observed.*** *If the sample has not lived through a crash, every path is optimistic and the ruin rate is a **floor, not an estimate**.*

The distinction that makes this layer worth having. There are two different "how much can I take out" questions, and the obvious one is not the one people mean:

- **Max sustainable** = the largest withdrawal whose ruin rate stays inside a tolerance. Its test only cares about not falling below a ruin threshold — so it will happily approve a path that consumes most of the account. On the reference book, a +555/month average produced a "sustainable" **2,966/month while the median account fell from 100k to 38k**. That is capital depletion on a schedule, not income.
- **Max preserving** = the largest withdrawal that still leaves the **median path with at least its starting capital** at the horizon. This is living off the income rather than eating the principal, and it is the number the cadence layer checks targets against.

Both are found by bisection and **rounded down, never to nearest** — a safety limit was chosen because it clears a threshold, and rounding it up can push the reported figure past the very threshold that selected it. A returned `0` is a real answer: *even withdrawing nothing breaches the tolerance*, i.e. the sample does not support drawing an income at all.

The tail budget sits deliberately alongside, not inside, the bootstrap. Because the resampling cannot invent a crash, the stress number is computed by **repricing the current book under a shock** — which does not depend on the sample having contained one — and is expressed in the only unit that makes an uncapped-tail book legible:

> *"A 20% move erases N months of average income."* Premium selling accumulates income linearly and gives it back in jumps. This is the size of one jump.

C — The guardrail: what a target does to behaviour when the month runs short

Phases A and B measure what the book earns and what it can support. Phase C watches the thing that destroys both: **the behaviour a monthly target induces when the month is behind.**

The failure mode is specific and predictable. Two thirds through the month, income is behind target, and the fastest way to close the gap is to sell more premium — bigger size, or strikes closer to spot. Both raise risk precisely when the motivation is a number rather than a setup. On a naked short book, that is how a good year ends.

So this layer refuses to make progress look easier than it is:

- **Premium sold is not income.** Credit collected on legs still open is an **obligation** — it can be handed back in full and then some. It is reported separately and **never added to booked P&L**. A trader "hitting target" on open premium has hit nothing.
- **The target is checked against measured capacity, not accepted as given.** If it exceeds the preserving withdrawal from layer B, the target is not ambitious, it is *unsupported* — and no amount of pace fixes that. This is raised as a critical warning regardless of how the month is going.
- **Risk headroom is checked before pace.** If the margin budget is already spent, the gap

cannot be closed within the plan at any pace, and the only honest answer is that the month ends where it ends.

- **Pace is expressed as a multiple of normal.** *"Needs 3.4× your usual daily rate with 9 days left"* is a statement about pressure; *"behind by 400"* is not. Past a stretch multiple the advice is to decide **now** that you will accept missing the target; past a chase multiple the warning names the mechanism — *"a number that far out of reach is how size creeps up and strikes drift toward spot: the trade gets taken for the target, not the setup."*

The verdict orders **pressure first, progress second** — a book that is nominally on pace but has already spent its margin budget reads as *chasing risk*, not as *on pace*.

3.23 Wheel cycles

A small store that links related legs — a cash-secured put and the covered call that follows assignment — into one **cycle**, so premium collected, net cost basis (`strike − total premium`, the break-even if assigned) and full P&L are tracked across the whole structure rather than per leg. Cycle statistics are **derived at query time** from the portfolio and closed ledgers rather than stored, so they cannot drift out of sync with the legs they describe.

3.24 Pin evidence — measuring the practitioner constructs

§3.11's gamma machinery (max pain, GEX, zero-gamma flip) is **practitioner folklore with a real mechanism underneath it** and no canonical academic definition. Rather than trusting it or discarding it, the system **measures it on this venue's own data**.

Every day, each live expiry gets one row: max pain, spot, the gap between them, net GEX, the gamma zone at spot. Once the expiry passes, the row sequence becomes a completed sample. Two tests, in a deliberate order of trustworthiness:

1. **Settlement evidence — the real test.** Where the expiry actually printed: did it land closer to max pain than it started, and did it settle within a tolerance of the pin?
2. **Convergence evidence — a proxy.** Did the spot-to-max-pain gap narrow between the first and the last pre-expiry capture?

Everything is scored on the **context at first capture** — the gamma zone, the approach direction — because that is what a forecast would actually have had available. Scoring on context known later would be a lookahead in miniature.

The honesty rules mirror the rest of the system, with one addition that matters:

- An expiry needs **≥2 captures** to measure convergence at all; a single row is dropped.
- **Daily expiries are segmented out and never silently pooled.** The pinning literature is about *serial* expirations carrying real open interest; crypto dailies carry very little, so pooling them with the monthlies drowns the signal in noise that looks like data.

- **The sample gate rises with the cut.** Segmenting splits the same evidence into thinner cells, so a segment must clear its own higher bar rather than inheriting the pooled minimum.

The advice layer consumes this through a **preference hierarchy** — settlement evidence over convergence, and within each, the most specific segment that clears its own bar, falling back to pooled. When nothing clears, the answer is `sufficient: false` **with no rate at all**, and callers are required to say the pin is **unmeasured** rather than quietly reverting to asserting that pins hold. That last rule is the whole point of the store: it converts a belief into either a number or an admission.

4. Security design

- **Encryption at rest.** Credentials and any bring-your-own API key are encrypted with a passphrase-derived key (scrypt over a per-value random salt → an authenticated symmetric cipher). The salt is embedded per value in a self-describing envelope, so there is no separate salt file to lose. A missing/wrong passphrase yields a clearly-reported **locked** state (configured but unusable) rather than a crash. Legacy plaintext is passed through and auto-migrated to ciphertext on the next save.
- **Optional login gate** for remote access: an HMAC-signed, HttpOnly session cookie with the signing key derived from the password; the whole app sits behind a device-level VPN, and the app password is defense-in-depth on top.
- **Secret hygiene is the real leak guard.** Credentials, the account directory, the bring-your-own-key file, and the market-history files are all excluded from version control. Encryption is defense-in-depth; the ignore rules are the primary control. (See 06 — Lessons for the incident that taught this the hard way.)

5. Risk constitution (the naked-strangle lens)

The reference book runs **fully naked short strangles on BTC** — an uncapped-tail posture. That shapes the risk view:

- Every "looks fine" reading is stress-tested against the tail, never just the mark. The executable-close and Capture% work exists precisely so a growing tail can't hide behind an optimistic mark.
- The backtest finding that crypto tails run far fatter than Gaussian is treated as a standing warning, not a curiosity — it argues directly against long-dated naked puts at wide-but-not-wide-enough strikes.
- Defined-risk structures are structurally favored in scoring, and the calibration loop can only *tilt*, never *override*, the market-signal and risk-limit gates.

02 — Implementation

How the system is built — described at the level of responsibilities, algorithms, and patterns. No source code; this is the "how it's put together" companion to the design.

1. Module map (responsibilities, not code)

The backend is ~45 small modules. The split that matters is **pure core** (math, testable offline) versus **I/O shell** (network, disk, endpoints).

Pure core — market & instrument math

Module	Responsibility
Chain builder	Merge instruments + tickers into unified contract objects; DTE filter; short cache
RV / IV stats	ATR, annualized realized vol, IV rank, IV/RV ratio
Volatility models	Parkinson / Garman-Klass / Yang-Zhang RV, HAR forecast, VRP, expected move, trend filter, funding summary
Fees	Trading-fee (with the venue's premium cap) and delivery-fee model; round-trip fee helper
Executable pricing	The single source of truth for "which side of the book do I get?" — bid on shorts, ask on longs, fee per fill; returns the mid / exec / after-fee decomposition and flags estimated quotes. Shared by the screener and the roll analyzer
Risk / margin	Per-leg OTM-scan initial margin; position sizing; capital & utilization limits; vol-target scale
Payoff	Piecewise-linear breakeven / max-profit / max-loss / reward:risk for any leg set
P&L calculator	Side-aware gross minus round-trip fees; multi-leg totals; fee-adjusted breakeven via bisection

Pure core — synthesis & judgment

Module	Responsibility
Strategy generators	~11 structure builders, each producing candidate legs from a chain
Strategy engine	Runs all generators across the chain (calendars handled outside the per-expiry loop)
Playbook	Transparent additive scoring of every strategy; ranked output; "Wait" logic; gamma-flip
Idea engine	Re-runs the engine filtered to the coach's own strategy/DTE/delta band; sizes candidates
Roll engine	Same-strike vs strike-adjusted roll tables for flagged legs, priced executable with both fees; per-day and per-day-yield figures
Roll score	Five-axis roll assessment (carry / VRP / safety / gamma / capital) with renormalized weights, full component breakdown, and gates kept separate from the score
Roll chains	Links legs by roll chain; martingale warnings; never-rolled vs rolled-in-profit vs rolled-tested evidence buckets, sample-gated
Theta edge	Per-leg breakeven move from θ and Γ, compared against realized movement; stale-greeks detector; θ/margin, vol-spike fragility, days-to-collect

Module	Responsibility
Coach	Net signed book greeks; book assessment; prioritized position actions; setup annotation
Analytics	Closed-trade win-rate / expectancy / profit factor, segmented; discipline metrics
Calibration	Personal-edge nudges from closed trades, regime-aware, expectancy-gated, capped
Scenario	Reprice the book under spot/IV/time shocks (Black-Scholes); P&L, margin, breakeven breaches
Expiry focus	Per-held-expiry max-pain / GEX; gamma-zone tagging of your strikes
Carry	Dated-futures basis vs funding vs VRP; rule-based income verdict
Cashflow	Monthly realized ledger net of fees; return on margin-at-entry; consistency (months-erased-by-worst); realized equity curve and drawdown
Sustain	Seeded bootstrap of monthly returns; max-sustainable vs max-preserving withdrawal by bisection; tail budget in months of income
Cadence	Month progress with open premium kept separate from booked P&L; target-vs-capacity and margin-headroom checks; pace as a multiple of normal
Backtests	Walk-forward RV studies and IV-rule studies; no lookahead

I/O shell

Module	Responsibility
Public exchange client	Instruments, tickers, candles, linear index & funding
Private exchange client	HMAC-signed read-only positions & wallet balance
Reference client	External venue's DVOL index + option book summary; delta reconstructed locally
Portfolio store	Leg CRUD; close (carries entry-context into the closed ledger); volume summary
Accounts store	Sub-account management; per-account directories; credential load/save
Regime history	Daily market snapshots; percentile / movement context
IV history	Intraday session-keyed IV snapshots; own-range percentile; seasonality
Pin history	Daily max-pain / GEX / gap rows per (underlying, expiry); settlement and convergence reports, segmented; the pin-prior lookup with its evidence hierarchy
Cycles store	Wheel cycles linking a CSP to its follow-on CC; stats derived at query time, never stored
Crypto	Passphrase-based encryption at rest
Auth	Optional login gate
App / endpoints	~40 routes wiring the above together

2. Key algorithms in prose

2.1 Assembling the chain

Fetch instruments and tickers separately, key both by symbol, and merge into one contract per symbol carrying strike, expiry, type, mark, mark-IV, bid/ask, and greeks. Compute days-to-expiry from the expiry timestamp (options settle at 08:00 UTC). Filter by a DTE window and cache briefly so a burst of endpoints doesn't refetch.

2.2 Computing the intelligence snapshot

For one underlying: pull the chain and spot (from the perp index, never the option ticker's underlying field), compute ATM IV at the reference expiry, 25-delta skew, IV rank, term structure, put/call ratio, the volatility-model block (RV, forecast, VRP, expected move, trend, funding), and the gamma/max-pain/zero-flip block. Everything is assembled into one summary object that the playbook, the coach, and the snapshot loop all consume. Optional inputs (funding, the account's closed trades for calibration) are passed in so the same pure function serves market-only and personalized callers.

2.3 Scoring the playbook

Start each strategy at 50. For each signal, look up that strategy's weight for that signal and add `weight × signal_strength`, appending a reason string with the points inline. Apply the defined-risk tie-break and the optional personal-edge nudges (after signal scoring, before the clamp). Clamp, sort, and emit the top pick in the legacy recommendation shape (so existing consumers don't break) plus the full ranked list. If the top score is below the action threshold, the recommendation becomes "Wait" but the ranked list still ships.

2.4 Fee-adjusted breakeven by bisection

The net-at-expiry P&L as a function of the terminal underlying price is monotonic on each side of the strike but kinked by the delivery-fee `min(...)`. Rather than solve the kink analytically, bracket the breakeven and bisect against the *exact* fee model until the net crosses zero. This automatically handles whichever fee term binds, and it agrees to the cent with the fee model used on real closes (same code path, different caller).

2.5 Personal-edge nudge

Attribute each closed trade to a strategy family by its short leg. Within the family, filter to a ±20 IV-rank band around the current regime if enough trades exist. Compute win-rate and average expectancy. Convert to a point nudge that is gated on positive expectancy and capped in magnitude, and attach the actual record as the reason. Below the minimum trade count, emit nothing.

2.6 Walk-forward backtest slicing

Iterate day *t* over the history. The "known world" at *t* is the slice up to and including *t*; the forward outcome is measured on strictly later bars over the study horizon. IV-rank percentiles are computed only over the trailing window of *past* data. Any accidental use of a future bar would inflate results, so the slice boundaries are the invariant the tests guard.

2.7 Pricing a structure the way it fills

Take a list of `(contract, side, quantity)`. For each leg pick the side of the book the fill actually takes — bid to sell, ask to buy — falling back to mid, then to mark, when the needed side is unquoted, and **record that the fallback happened**. Compute the leg's fee from the index price and the executable premium. Accumulate three running totals with the sign convention *positive = credit received*: at mid, at executable prices, and executable minus fees. Return all three plus their difference decomposed into spread cost and fees, the retention ratio, per-leg detail including a **fee-folded net price**, and two boolean flags: credit-vanishes and quote-estimated.

Two consequences of doing it this way rather than applying a haircut:

- The identity `mid − net = spread + fees` is **checkable**, and a test asserts it. A single "adjusted credit" would be unfalsifiable.
- Because each leg carries a fee-folded price, the payoff engine produces fee-inclusive breakevens **without importing the fee model** — fees are folded in at the one place that knows about them, and every other consumer stays ignorant of venue fee rules.

2.8 Building a roll table

For a short leg that is tested or near expiry: buy the old leg back at its **ask**, then for each candidate expiry (soonest first, capped) sell the new leg at its **bid**. Two tables are produced — same strike, and strike adjusted toward a defensive delta target. Each row pays a trading fee on both fills.

Per row, derive `added_days` from the DTE difference and divide the after-fee credit by it; divide again by collateral (strike for puts, spot for calls — the same convention the matrix tab's annualized yield uses) to get yield-per-day. **These per-day figures, not the raw credit, are what makes rows at different expiries comparable**, because the VRP term structure slopes down steeply enough that a bigger absolute credit often buys less premium per unit of time-risk.

Then score the row (§2.9) — which requires locating the leg *being closed* in the chain, for the gamma-relief axis. When it cannot be located the row still prices correctly and simply carries no score, rather than scoring on a guessed old leg.

2.9 Scoring a roll

Each axis produces `(raw value, subscore, weight, plain-English reason, display string)` and appends to a component list; the score is the weight-average over **whichever axes computed**, with the used-weight total reported so a partially-scored roll is visibly partial. Subscores come from piecewise-linear interpolation over fixed breakpoints, clamped at both ends.

Gates accumulate into a separate list. Two of them are *hard* — a debit roll, or a new contract whose IV is at or below forecast RV — and force the verdict to *questionable* regardless of the average; the rest are advisory and surface next to the score. The component list is always returned, so the UI can show a trader exactly which axis they are disagreeing with.

2.10 The theta-efficiency read

Convert position theta down to per-unit so the quantity cancels against per-unit gamma, then compute the breakeven move and express it as a percentage of spot. Compute the leg's own implied daily move and take the ratio: if it is far from the expected ≈0.96 the greeks are stale, so flag the leg and treat its read with suspicion. Divide the breakeven percentage by the **realized** daily percentage supplied by the caller (from ATR or an RV forecast) for the edge ratio and verdict. If the caller supplies no realized baseline, the verdict stays `unknown` — the system does **not** fall back to implied, because that is the circular comparison the module exists to avoid.

The book-level roll-up reports how many legs scored, how many are underpaid, how many have suspect greeks, and the median edge ratio — deliberately the median, since one deep-OTM leg with a near-zero gamma can produce an enormous ratio that a mean would not survive.

2.11 Searching for a withdrawal limit

Both withdrawal limits are found the same way, and the shape of the search is the interesting part. Start with a bracket `[0, hi]`; **double** `hi` **until it actually fails the test**, because a bracket whose upper end still passes would let bisection converge to the bracket edge and report a limit that was never tested. Then bisect a fixed number of times, always keeping the *passing* side as the lower bound. Finally **floor** the result to two decimals.

Before searching at all, test W=0. If withdrawing nothing already fails, return `0` with a note — that is the real answer, and a bisection over an empty feasible set would otherwise return a meaningless number.

The two tests differ only in the predicate: ruin rate within tolerance (sustainable) versus median terminal equity at or above starting capital (preserving). Reusing one search with a swappable predicate is what makes it obvious that they are the *same* procedure answering *different* questions — which is the point the numbers themselves make loudly.

2.12 Assembling the cadence check

Compute what the month has booked and, **separately**, the premium sold on legs opened this month that are still open. The second number is never added to the first anywhere in the pipeline; it is passed through to the UI as its own field with its own label.

Then run the checks in a fixed order, because the order encodes which objection dominates:

first whether the target exceeds measured capacity (a structural problem no month can fix), then whether margin headroom exists to add at all (if not, pace is moot), and only then pace as a multiple of the normal daily rate. Any input may be absent — average month, preserving withdrawal, margin figures all come from other layers — and a missing input **withholds its check** rather than substituting a default. The verdict reads pressure before progress: a critical warning outranks being nominally on pace.

2.13 Turning pin snapshots into evidence

Group rows by expiry; keep expiries strictly in the past that have at least two captures carrying a usable gap. For each, take the first and last capture: the gap at each end, whether it narrowed, and — critically — the **context at first capture** (gamma zone, net GEX, approach direction) as the "prediction" being scored. Classify the expiry as daily/weekly/monthly/quarterly from its calendar position rather than from a listing flag. Rates are then computed per segment with an n attached to every one, and the pin-prior lookup walks the hierarchy from most to least trustworthy, returning the first cell that clears its own minimum. The function's most important return value is the one where nothing clears: it returns *insufficient* rather than the pooled number, so a caller cannot accidentally quote thin evidence as if it were the measured rate.

2.14 Background snapshot loop

Wake on a coarse interval. For each underlying, check whether a daily regime snapshot and/or an intraday IV snapshot is *due* (by UTC day and by session window). If either is due, compute the intelligence snapshot once and write whichever stores are due. Window-based due-ness (not a fixed instant) means an arbitrary wake phase never misses a session and absorbs daylight-saving shifts in the reference session boundaries.

3. API surface (shape, not signatures)

Roughly 40 endpoints, grouped:

- **Market:** chain, strategy suggestions, the intelligence snapshot, matrix, arb scan, carry, the RV and IV backtests.
- **Portfolio:** enriched legs (with live mark, greeks, executable close quote, per-leg margin), volume summary, cycles, risk summary, analytics.
- **Income:** the monthly cash-flow ledger and equity curve, the sustainability report (bootstrap, both withdrawal limits, tail budget), and the cadence check against a target.
- **Coach:** the fused brief, scenario stress test, expiry focus, the roll analyzer (priced table + five-axis score) and the roll-chain evidence report, regime history, grounded natural-language Q&A.
- **Account & settings:** sub-account CRUD and switch, credential status, risk config, the

bring-your-own-key config.

- **Reconciliation:** the position preview/diff and the import.

Every account-scoped endpoint snapshots the active account directory before its first await.

Every endpoint that can hit the network soft-fails to a labeled "unavailable" rather than throwing, so one dead upstream never breaks a whole screen.

4. Frontend patterns

- **One file, tab-switched surfaces.** Each tab lazy-loads its data on first open and on an explicit refresh.
- **Event delegation on stable containers.** Editable tables (the calculator, the legs table) attach one handler to the container, not one per row, so a full re-render never moves the caret while the user is typing. Writes are **debounced**.
- **Render from synthetic payloads.** Render functions are driven purely by a data object, so they can be exercised in a browser with a fabricated payload — which is exactly how the UI is verified (below) without needing live positions in a particular state.
- **Grouping and collapsing for scale.** Once the book grows, flat lists become noise, so legs and the live-margin breakdown are grouped per underlying with subtotals, and dense per-position detail hides behind a toggle whose open/closed state survives auto-refresh.
- **Color encodes the decision, not the raw number.** Where a value's usefulness depends on interpretation (close-quote economics, capture, margin share), color follows the *decision* meaning, with the raw/secondary detail demoted to a tag.

5. Testing and verification

5.1 Unit tests against synthetic fixtures

Because the math core is pure, every model is tested offline:

- Volatility estimators are checked against **seeded geometric-Brownian-motion candles** with a known vol.
- Fees are checked against the exchange's own **worked examples**.
- The payoff engine is checked against **closed-form** payoffs of standard structures.
- The margin formula is checked against the exchange's **worked margin example**.
- Backtests are checked against **seeded ground truth** so the walk-forward slicing is provably lookahead-free.
- Calibration, scoring, and the IV/skew sign conventions each have targeted tests — including a regression test that would catch the skew-sign bug.
- Invariants that protect the JSON layer (never emit `inf`/`nan`) are asserted directly.

5.2 Browser-driven verification of the UI

Beyond unit tests, UI changes are verified by driving a real browser against the running app: render the affected surface (often from a **synthetic multi-asset payload** so edge cases appear deterministically), read the resulting DOM back, and assert on the actual rendered values and colors. Console and network are checked for errors. This catches the class of bug that unit tests can't — a formula that's right but wired to the wrong column, or a color rule that contradicts the number next to it.

5.3 The verification mindset

A change isn't "done" because the code looks right; it's done when the affected flow has been exercised and observed. Several of the most important fixes in the project (the surface-avg-vs-ATM IV bug, the close-quote color confusion) were found precisely because a rendered result was read back and compared against what the number *should* mean.

03 — Usage

How a trader actually drives the system. Organized as the workflow first, then each surface, then the decisions each one supports.

1. The daily workflow

1. **Open on the Portfolio.** The landing surface is the live book, not a screener — because the first question every session is "what do I already have, and does anything need attention?" Legs are grouped per asset with margin, unrealized P&L, and theta subtotals, ordered so the heaviest-margin asset is on top (the trim shortlist).
2. **Clear any sync drift.** If the local book has drifted from the exchange (a fill, a partial close, an early close), a banner surfaces it with one-click import/update or record-close. It refreshes on every reload so it reflects reality immediately.
3. **Read the regime.** The Intelligence/Coach surfaces answer "is vol rich, which way is the tape leaning, where's the gamma?" in one named regime line plus supporting signal cards with percentile and 5-day-movement context.
4. **Check the coach brief.** The fused view lists prioritized position actions (defend / near-expiry / roll-window legs) and new-setup ideas already annotated by how they fit the current book.
5. **Act, then record.** Execute manually on the exchange, then import or add the leg so the entry-context flywheel captures the market state at entry.
6. **Review the edge periodically.** The scorecard shows where your realized results actually come from — by strategy, underlying, DTE-at-entry, and IV-rank-at-entry.

2. The surfaces

2.1 Portfolio

The operational center. Per-leg it shows: side, strike, expiry with a DTE pill, quantity, entry, mark, the **executable close quote** (ask for shorts, bid for longs) with a spread / one-sided tag, **Capture%** (premium actually kept if closed now at that quote), profit% (on mark), theta/day, unrealized P&L, **per-leg margin** with its share of book margin, and probability-of-profit / breakeven distance.

Decisions it supports:

- *Which legs to trim* — sort by margin share; the biggest consumers are the shortlist.
- *Which shorts to buy back cheaply* — Capture% ≥ a take-profit threshold, close-quote color green, means locking in most of the credit is available now.
- *Which legs are stuck* — a one-sided or very wide close quote means you can't exit at market; use a limit.

- *Whether a "green on mark" leg is actually a loser to close* — Capture% negative says yes.

A **trade-volume strip** (premium transacted, contracts, notional) and a **collapsible live-margin panel** (grouped per underlying, sourced from the exchange's wallet balance) sit above/around the table.

2.2 Screener

Runs the strategy library across the chain and lists liquid candidates with net credit, breakevens, probability-of-profit, annualized yield, and the exact payoff KPIs (max profit, max risk, reward:risk). The playbook can deep-link here with the right strategy preselected.

2.3 Intelligence

The market read. IV rank, ATM IV (with surface-average demoted to a sub-value so they're never confused), 25-delta skew, term structure, VRP, expected-move-vs-realized table, trend, funding, gamma flip and max pain. A **carry monitor** compares dated-futures basis, funding, and VRP. An **external-reference card** shows the deepest-venue DVOL/ATM/skew with explicit "Bybit − reference" richness gaps — labeled as reference, not signal. **IV-history** and **backtest** panels show accrued own-venue history and the walk-forward studies.

2.4 Coach

The synthesis surface:

- **Trade brief** — regime cards, net signed book greeks, prioritized position actions, and new-setup ideas annotated by book-fit (with concrete, sized strikes).
- **Scorecard** — realized win-rate/expectancy/profit-factor segmented four ways, plus discipline metrics (premium-capture efficiency, loss/win size asymmetry), all with sample-size-guarded insights.
- **Scenario stress test** — three sliders (spot shock / IV shift / days forward); dial all three, then run, and see book P&L impact, projected margin/utilization with a warning banner, and newly-breached legs.
- **Expiry-week focus** — per-held-expiry max-pain/GEX with each of your strikes tagged long- or short-gamma.
- **Roll analyzer** — for flagged legs, a comparison of same-strike vs strike-adjusted rolls with net credit and resulting delta (a table, not one opaque "best" pick).
- **Ask the coach** — a grounded natural-language Q&A that answers only from the on-screen brief/scorecard data, with a strict "never invent numbers, advisory-only" prompt and a user-set daily question cap.

2.5 Calculator

A multi-leg P&L what-if. "Load my open book" opens an expiry-grouped picker that prefills legs from the live book (buy-back = current mark, index = live spot). Per leg you set taker/maker and optionally strike + type to get a **fee-adjusted breakeven**; per-leg and whole-position net-after-fees update live. A "daily option" flag drops the delivery-fee term. `close = 0` models expire-worthless. It uses the *same* fee model as real closes, so the what-if agrees with reality to the cent.

2.6 Settings

Sub-account switch, API credential status (with a locked-state badge when encryption is on but the passphrase is missing), risk configuration (capital, max utilization, vol-target), and the bring-your-own-key config for the natural-language Q&A with its daily budget.

3. Auto-refresh and reconciliation

The portfolio auto-refreshes on a short default interval (adjustable, or off). Every refresh re-diffs the book against the exchange and updates the sync banner. Because the diff runs inside the same reload path as every mutation (import, add, close, expire, delete, account switch), any action clears or updates the banner immediately rather than leaving a stale alert until the next timer tick.

4. What the system will and won't do for you

Will: assemble the full market read, score structures transparently, size to your limits, show you the real (executable, fee-inclusive) cost of getting in and out, flag the legs that need attention, remember your entry context, and tell you where your realized edge actually is — hedging every claim by how much data backs it.

Won't: place, modify, or cancel any order; move funds; give you a number it can't support; or hide a growing tail behind an optimistic mark. Execution, and the judgment of whether to execute, stay with you.

04 — Workflows

The concrete, repeatable procedures the system is built around. Each is a sequence of steps mapping a decision to the surfaces and signals that support it. The design goal is that none of these requires holding state in your head — the tool surfaces the next input at each step.

1. The daily desk review

The opening ritual every session.

1. **Land on the Portfolio.** See the whole book grouped per asset with margin, unrealized P&L, and theta subtotals; the heaviest-margin asset sits on top.
2. **Clear sync drift.** If the book has diverged from the exchange, resolve the banner first — import new fills, update partial closes, record early closes — so every later number is computed against reality.
3. **Read the regime line.** One named banner ("High-IV, range-bound, contango, premium-rich") plus signal cards with percentile and 5-day-movement context.
4. **Scan the action items.** The coach brief lists prioritized position actions (defend / near-expiry / roll-window) and net signed book greeks.
5. **Check margin headroom.** The live-margin panel (from the exchange wallet) shows IM utilization and the buffer above maintenance — the ceiling on any new risk.
6. **Note anything for later.** Legs approaching a take-profit Capture%, legs whose close quote has gone one-sided, expiries clustering.

2. Is vol rich? — the regime assessment

Before selling any premium, decide whether premium is worth selling *now*.

1. **VRP first.** Implied minus forecast realized vol. Positive and wide = premium is rich; thin or negative = the seller's edge isn't there.
2. **IV rank** for context — where current IV sits in its own recent history (a true percentile once history has accrued).
3. **Skew** — 25-delta put-minus-call. Rich puts (positive) reward put-side selling and skew-harvest structures.
4. **Term structure** — contango favors calendars; backwardation warns of near-term stress.
5. **Expected-move-vs-realized** — is the market pricing more movement than tends to happen?
6. **External reference** — the deepest-venue DVOL/ATM/skew as a labeled sanity check, never as the deciding signal.
7. **Backtest context** — remember the standing finding: ATM VRP is thin, so the edge lives in the OTM skew/tail, not ATM richness.

3. Entering a new position

1. **Consult the playbook.** It ranks every structure for the current regime with visible score components and reasons; the top pick fits the regime, or it says "Wait."

2. **Respect the personal-edge nudge.** If your own history in similar regimes tilts a family up or down, that shows as a reason line on the ranked card.

3. **Deep-link to the screener** with that strategy preselected; pick concrete, liquid strikes with acceptable credit / breakevens / probability-of-profit / annualized yield.

Read the credit as the executable one — shorts fill on the bid, longs on the ask, and every leg pays a fee. Check **edge retention**: what fraction of the mid credit actually survives the order book. A candidate retaining 20% is mostly a gift to the market maker, and the two flags to respect are *credit vanishes* (mid says credit, the book says debit) and *quote estimated* (the side you need is unquoted, so the price shown is a model price — use a limit order and expect to negotiate).

4. **Check the payoff shape** (KPIs: max profit, max risk, reward:risk) — and the diagram family in 05 — Strategy payoffs — so the risk profile is what you intend (especially: is the tail bounded?).

5. **Size to limits.** The contracts-allowed figure comes from your capital, max-utilization setting, and vol-target scale. A `0` means your limits don't allow it here.

6. **Stress-test if adding meaningful risk** (workflow 8).
7. **Execute manually on the exchange.**
8. **Record the leg** (import or add) so the entry-context flywheel snapshots the full market state at entry — the fuel for future calibration.

4. Managing an open position

1. **Watch the action items.** A leg gets flagged when it needs defending, nears expiry, or enters the roll window.

2. **Read the executable close, not the mark.** The close-quote column shows the price you'd actually transact and its Capture% (premium kept if you close now).

3. **Decide:**
   - *Take profit* → workflow 6, when Capture% clears your threshold.
   - *Roll* → workflow 5, when the thesis holds but time/strike needs adjusting.
   - *Defend* → tighten or hedge a threatened side; re-check net book greeks after.
   - *Hold* → nothing to do; theta accrues.
4. **Re-check margin and net greeks** after any change.

5. Rolling a position

1. **Open the roll analyzer** for the flagged leg.
2. **Compare the two paths it tables**, soonest-expiry-first: **same-strike** (buy time only) and **strike-adjusted** (move further OTM toward a defensive delta target, buy time).

3. **Compare on credit *per day*, not on the headline credit.** The VRP term structure slopes down steeply, so rolling further out for a bigger absolute credit frequently buys *less* premium per unit of time-risk. The per-day and yield-per-day columns exist because the raw number is not comparable across expiries.
4. **Read the credit as executable.** A roll crosses the spread **twice** and pays a fee on both fills. The *mid flatters* flag marks the case where mid calls it a credit and the real book calls it a debit.
5. **Read the five-axis score — and read the axes, not just the number.** Carry, VRP on the contract you're *opening*, safety in expected moves, gamma relief, margin change. Two things to internalize:
   – a **same-strike roll scores worse on safety as tenor grows**, and that is correct — it buys time, not safety, because the move budget grows with $\sqrt{T}$;
   – the **VRP axis prices the new leg**. Rolling into cheap vol to repair an old position is the classic losing roll, and nothing else on the screen will show it to you.
6. **Treat gates as vetoes, not as inputs.** A debit roll or one selling below forecast RV caps the verdict at *questionable* however well it averages. A new strike inside one expected move is an open gate even on a high score.
7. **Check the chain before rolling again.** If this leg has already been rolled, look at the chain warnings: rolled ≥3×, size growing, cumulative P&L negative, every roll made while losing. That pattern is how an uncapped-tail book dies, and each individual roll looks like "collect more credit" right up until it doesn't.
8. **Prefer a roll that stays a credit** and moves delta toward neutral, unless you're deliberately taking a directional view.
9. **Execute manually**, then update the book (which re-runs the sync check). The roll context — *were you in profit or defending?* — is captured at that moment and cannot be reconstructed later; it is what makes the weekly review's "does rolling work for me" answerable at all.

6. Taking profit / closing

1. **Confirm Capture%**, not mark-profit. Capture% is the premium actually kept at the executable quote; the gap to mark-profit is the cost of reality.
2. **Check the close quote is two-sided.** A "one-sided" tag or a very wide half-spread means you can't exit cleanly at market — use a limit.
3. **For a short-premium book**, closing when the buy-back costs a small fraction of the credit locks in most of the max profit; a deep-OTM leg near worthless may be left to expire (OTM expiry pays no delivery fee).
4. **Model it first if unsure** in the calculator (workflow below), which nets both trading fees and, on exercise, the delivery fee.

5. **Record the close** so realized P&L (net of fees) feeds the scorecard and calibration.

7. Reconciling the book (sync)

1. **The diff runs automatically** on every portfolio reload and on the short auto-refresh.
2. **Resolve each drift type:** *new* (import the leg), *size-changed* (update via import — a partial close or add), *closed-early* (record the close to book realized P&L).
3. **Because the diff runs inside the reload path**, acting on a drift clears the banner immediately rather than waiting for the next timer tick.

8. Scenario stress test (before events / before adding risk)

1. **Dial three inputs:** spot shock, IV shift, days forward (or a preset: selloff, crash, rally, time-only).
2. **Run** — the whole book is repriced under the shock (Black-Scholes on matched legs).
3. **Read the outputs:** book P&L impact, projected margin and utilization with a warning banner if it breaches, and any newly-breached breakevens.
4. **Note skipped legs** (unrepriceable / no live IV) reported separately so the totals are never silently partial.
5. **Decide** whether the post-shock margin and tail are acceptable *before* adding the position.

9. The P&L / breakeven what-if (calculator)

1. **Load the open book** into the multi-leg calculator (expiry-grouped picker; prefilled at current mark and live spot).
2. **Set taker/maker** per leg and, for a breakeven, the strike + type.
3. **Read per-leg and whole-position net after fees**, and the **fee-adjusted breakeven** (solved against the exact fee model; a "daily option" flag drops the delivery term).
4. Use `close = 0` to model expire-worthless. The numbers agree with a real close to the cent because it's the same fee model.

10. Weekly edge review

1. **Open the scorecard.** Realized win-rate, expectancy, and profit factor, segmented by strategy, underlying, DTE-at-entry, and IV-rank-at-entry.
2. **Read the discipline metrics:** premium-capture efficiency and loss/win size asymmetry.
3. **Read the roll-chain evidence.** Completed chains you never rolled, rolled while in profit, and rolled while tested, compared on average P&L. This is the only honest answer to *"does rolling work for me"* — and note that a chain containing both kinds of roll counts as **defensive**, because the defensive roll is the risk-relevant event.
4. **Read the theta-efficiency roll-up.** How many open legs are *underpaid* — realized movement running above what their theta pays for — plus the median edge ratio and any legs

flagged for stale greeks. A book can print positive theta every day and still be losing in expectation; this is the number that says so.

5. **Trust only what the sample supports** — insights are gated on size and say "only N so far" below it.
6. **Let it feed forward.** These same outcomes drive the calibration nudges that tilt next week's playbook toward what has actually worked for you (capped so market signals still dominate).

11. The monthly income review

Run once a month, after the month closes. This is the workflow that decides whether the book is an *income stream* or merely a profitable strategy — they are not the same thing, and only this one can tell them apart.

1. **Read the completed months, never the running one.** The in-progress month is shown but excluded from every statistic; a partial month flatters or damns the record at random.

Below the minimum number of complete months, no averages are quoted at all — a hundred trades inside three months is still three data points.

2. **Look at months-erased-by-worst before the average.** How many average *winning* months the worst month wiped out. On a negatively-skewed short-vol book this ratio, not the mean, decides whether the income is spendable.
3. **Read return on margin, not just P&L.** 500 earned on 5,000 of margin and 500 on 50,000 are different businesses. Check the coverage figure too — trades opened before entry-context capture have no recorded index, so their margin is left out rather than guessed.
4. **Remember what the equity curve is.** Realized P&L only. Open positions are not marked, so a smooth line here is entirely compatible with a large unrealized loss you have not taken.
5. **Read the preserving withdrawal, not the sustainable one.** The "sustainable" figure only promises you will not hit the ruin floor — it will happily approve a withdrawal that consumes most of the account on the way. The preserving figure is the one that leaves the median path with its capital intact. If you want a number to live on, that is the number.
6. **Treat the ruin rate as a floor.** The bootstrap resamples the months you have actually lived through, so **it cannot produce a month worse than your worst**. If your sample has no crash in it, every path is optimistic.
7. **Read the tail budget instead, for the tail.** A repriced shock on the *current* book, expressed as months of average income. This one does not need your sample to have contained a crash — which is exactly why it exists next to the bootstrap rather than inside it.
8. **Set next month's target against capacity, not against ambition.** If the target exceeds the preserving withdrawal, hitting it consistently means drawing down principal, whatever any individual month does.

12. Mid-month: the cadence check

1. **Booked P&L is progress. Premium sold is not.** Open credit is an obligation that can be handed back in full and then some; it is reported separately and never added.
2. **Check headroom before pace.** If the margin budget is already spent, the gap cannot be closed within the plan at any pace, and the month ends where it ends.
3. **Read pace as a multiple of your normal daily rate.** "3.4× your usual rate with 9 days left" is a statement about pressure; "behind by 400" is not.
4. **Past the stretch multiple, decide now that you will accept missing the target.** Deciding in advance is the whole point — the failure mode is not a bad trade, it is a trade taken for the target rather than for the setup, and it arrives as size creeping up and strikes drifting toward spot.

13. Asking the desk advisor

A book-aware assistant workflow for judgment questions ("should I roll this?", "is this vol cheap?", "size this", "where's my edge?"):

1. It reads the live book and market data first (API, with local files as a fallback).
2. It answers **grounded only in that data** — never inventing numbers — and stays advisory: it enforces the VRP gate, the defined-risk bias, your risk-limit caps, and behavioral discipline drawn from your own closed trades.
3. It **never places an order.** Every suggestion is yours to execute or ignore.

05 — Strategy payoffs

The structures the playbook scores and recommends, each with its expiry-payoff diagram.

How to read these. The horizontal axis is the underlying price at expiry; the vertical axis is profit/loss per one unit of underlying. Green is profit, red is loss, the bold grey line is zero, and the open circles are breakevens. The strikes are marked on the price axis.

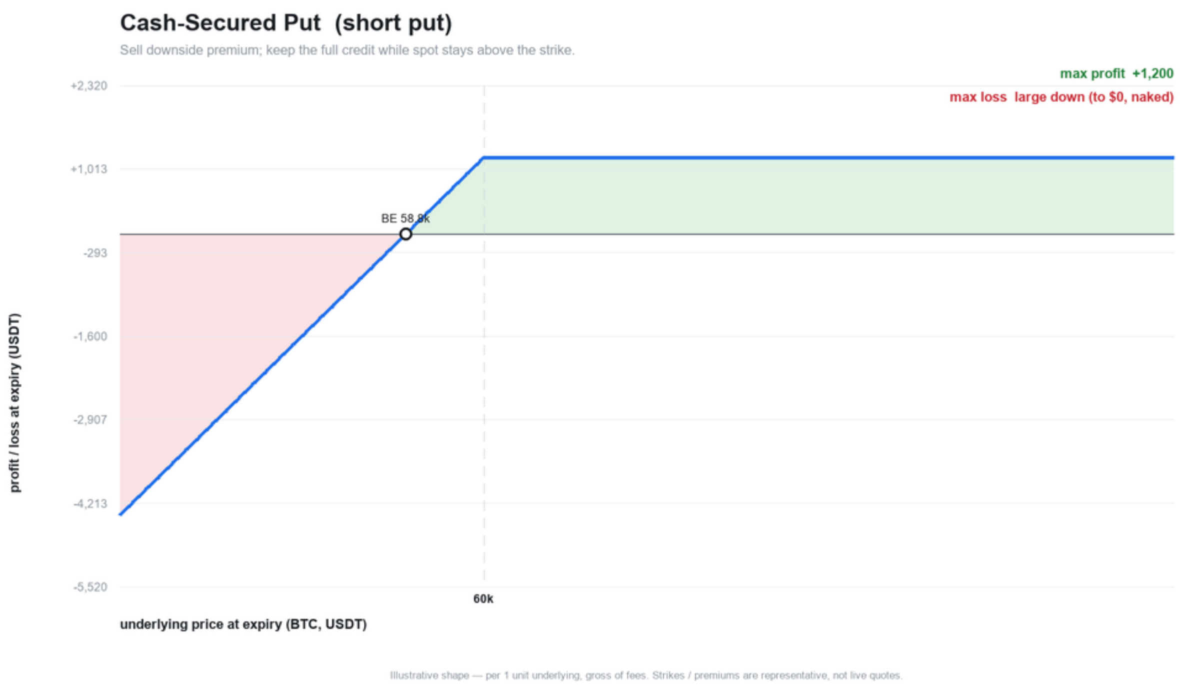
All shapes are **illustrative and gross of fees** with representative BTC-scale strikes and premiums — not live quotes. The real system computes exact breakevens and max profit/loss from the live chain and applies the fee model on top; see 01 — Design §3.10 for the payoff engine and §3.1 for fees.

A single rule organizes the whole library: **defined-risk structures are structurally favored in scoring**, and any structure with a naked tail is flagged as such. Read every diagram tail-first — *where does the loss stop?*

Neutral premium-selling (the core income structures)

Cash-secured put

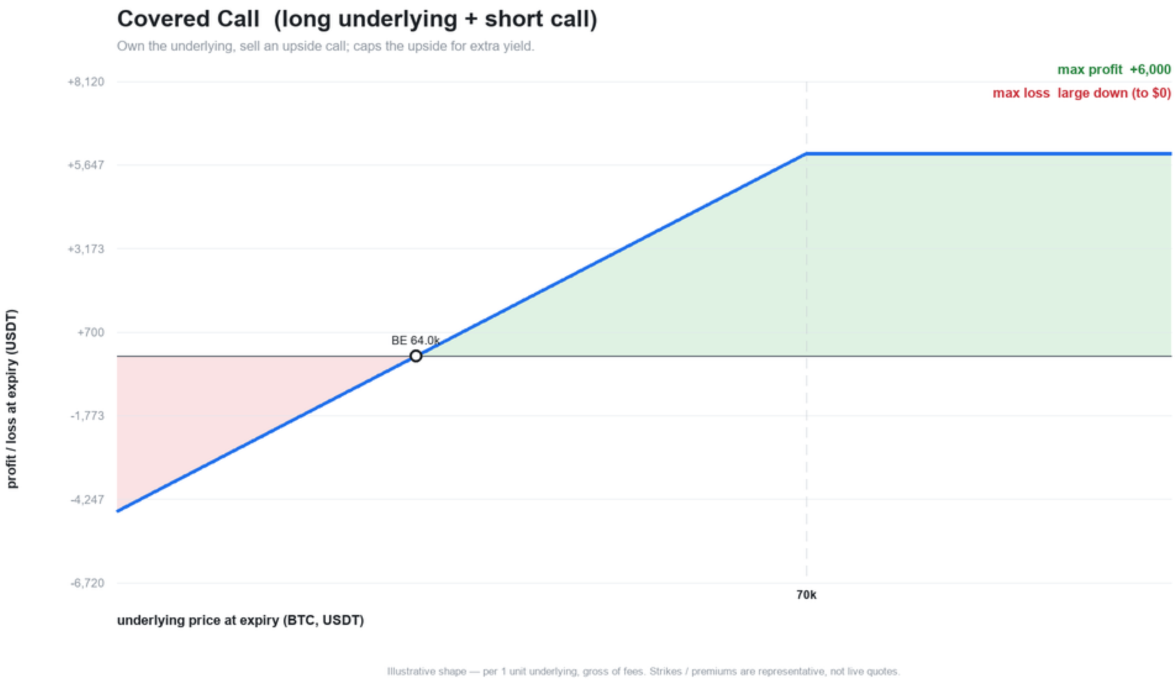
Sell an out-of-the-money put; keep the full credit while spot holds above the strike. The downside is the naked put tail (bounded only at zero). Picked in neutral-to-bullish, high-VRP regimes.



Cash-secured put payoff

Covered call

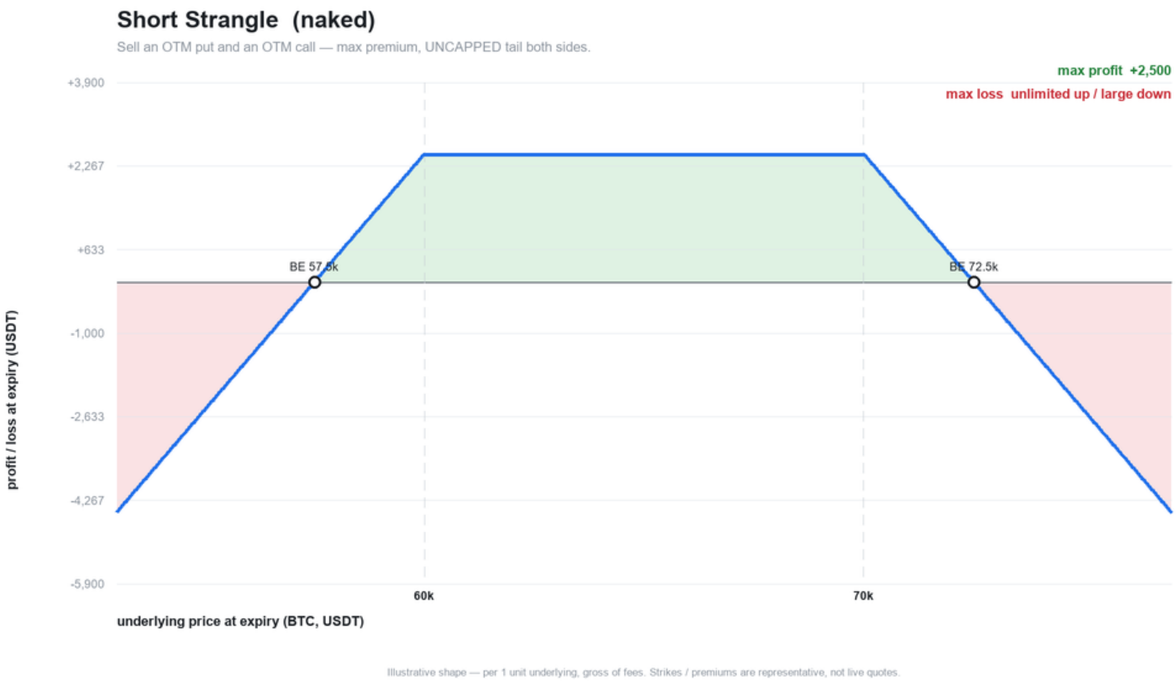
Own the underlying and sell an upside call for yield; the call caps the upside above its strike while you keep full downside participation. Picked when holding the underlying in a neutral-to-mildly-bullish regime.



Covered call payoff

Short strangle (naked)

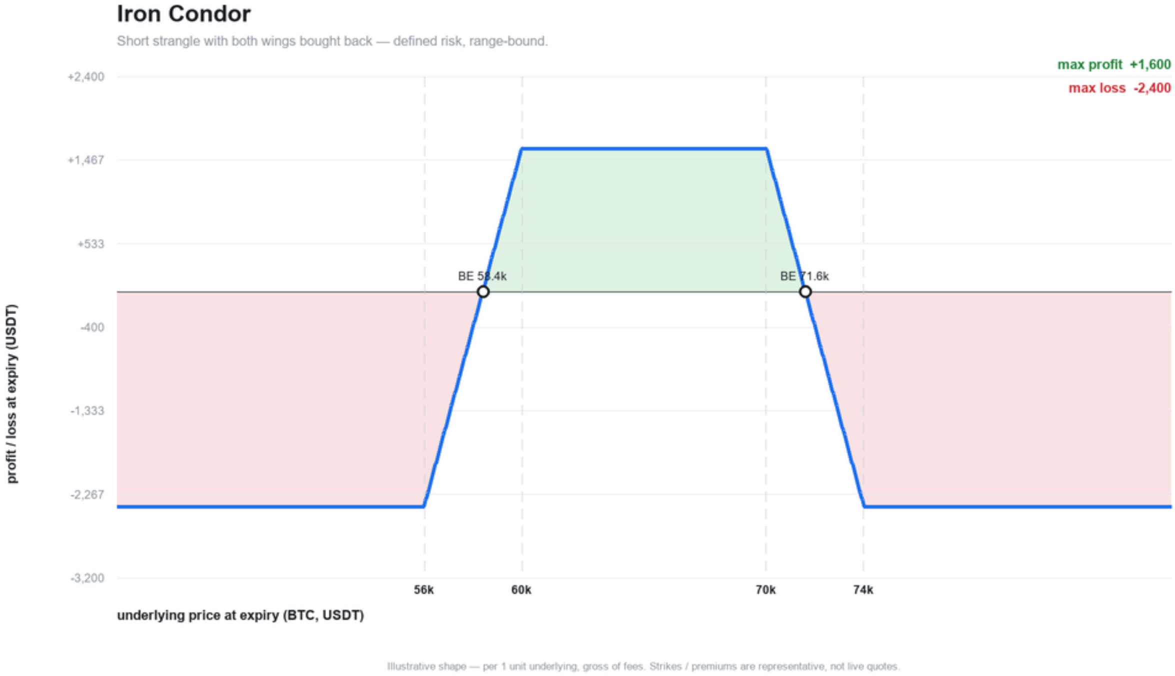
Sell an OTM put and an OTM call for maximum premium. The profit is capped at the credit between the strikes; **both tails are uncapped**. This is the reference book's core posture — and exactly why the executable-close and Capture% tooling exists (a growing tail must never hide behind an optimistic mark). Picked in high-VRP, range-bound regimes; carries a structural scoring penalty for its naked tails.



Short strangle payoff

Iron condor

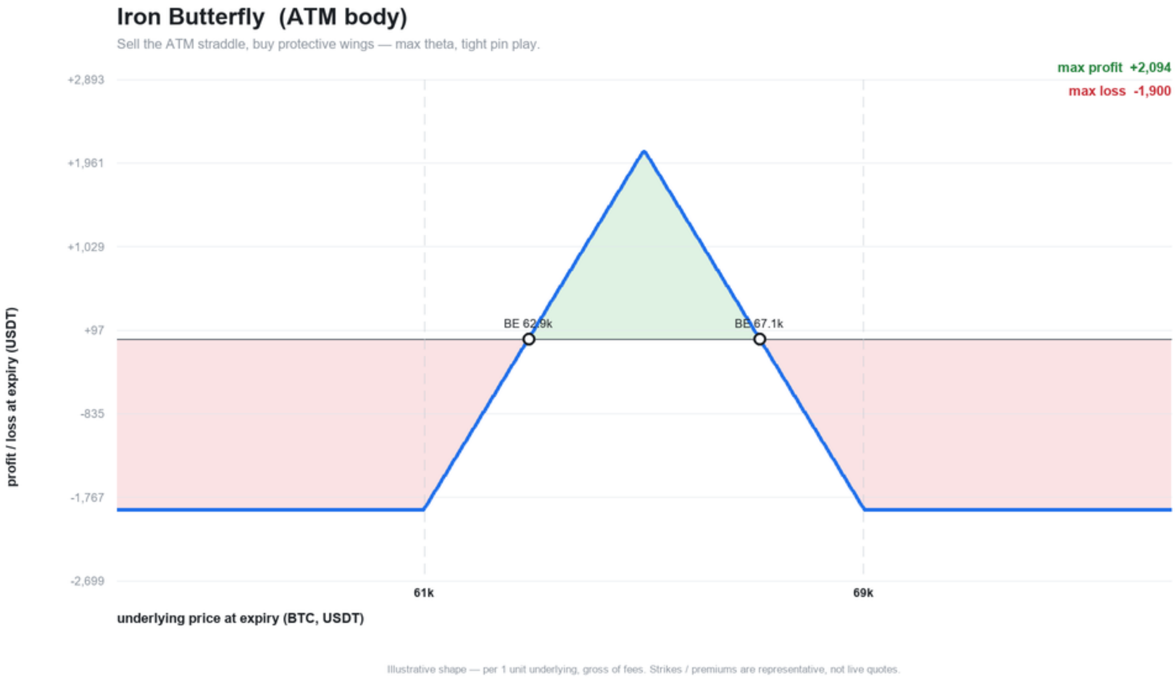
A short strangle with both wings bought back — the same range-bound thesis with **defined risk**. Lower max profit than the strangle, but the tails are capped. Favored over the strangle at equal signal edge.



Iron condor payoff

Iron butterfly

Sell the at-the-money straddle and buy protective wings — maximum theta, a tight pin play around the current price. Defined risk. Picked when expected move is rich and the tape is pinned.

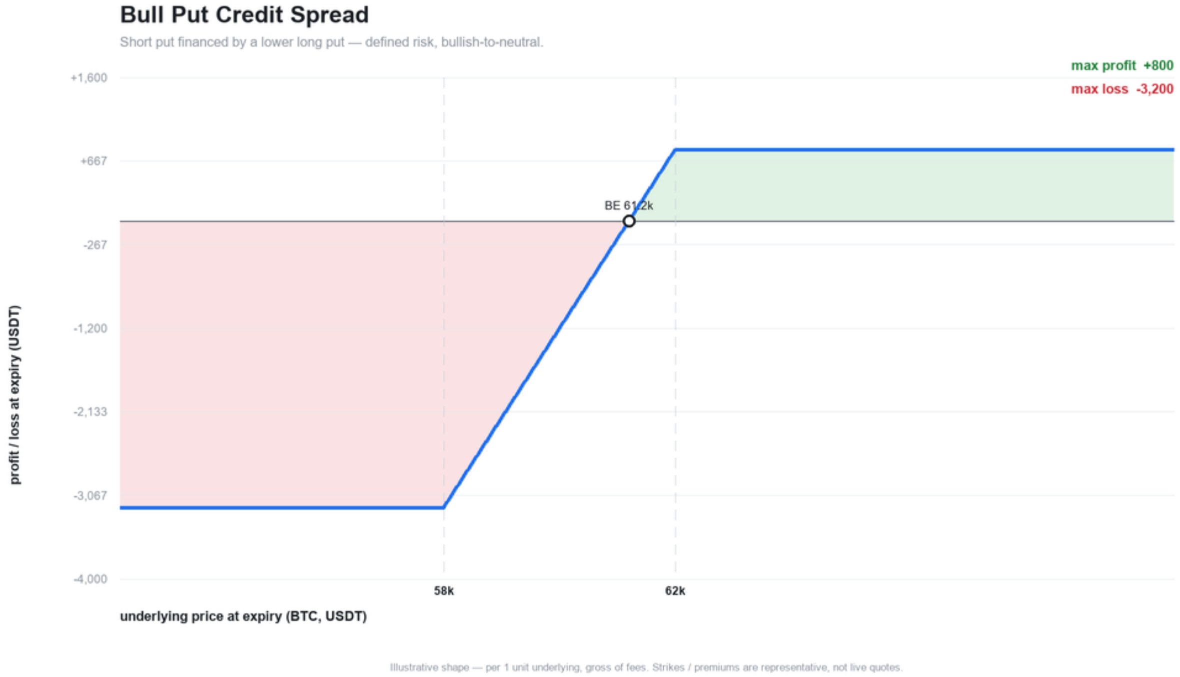


Iron butterfly payoff

Defined-risk directional

Bull put credit spread

A short put financed by a lower long put — a bullish-to-neutral credit trade with a hard floor on the loss. One of the "finally reachable" defined-risk credit structures. Picked when the trend filter leans up and VRP supports selling.



Bull put credit spread payoff

Bear call credit spread

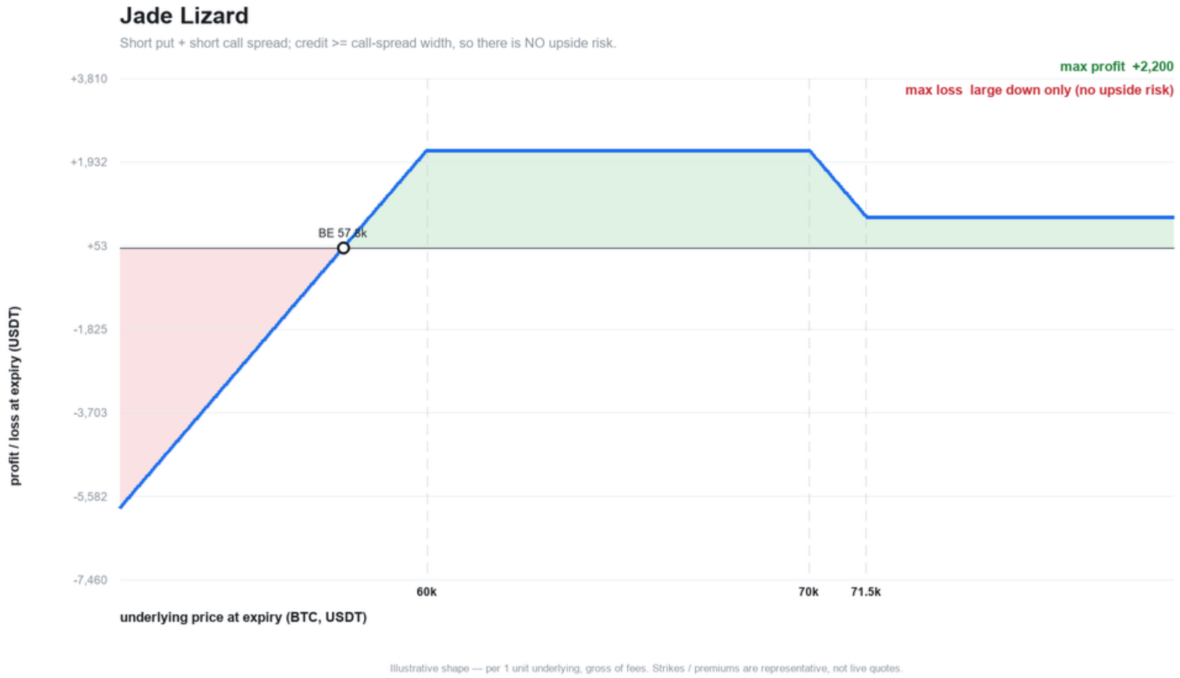
The mirror: a short call capped by a higher long call, bearish-to-neutral, defined risk. Picked when the trend leans down and call premium is worth selling.

Bear call credit spread payoff

Skew and tail harvesters

Jade lizard

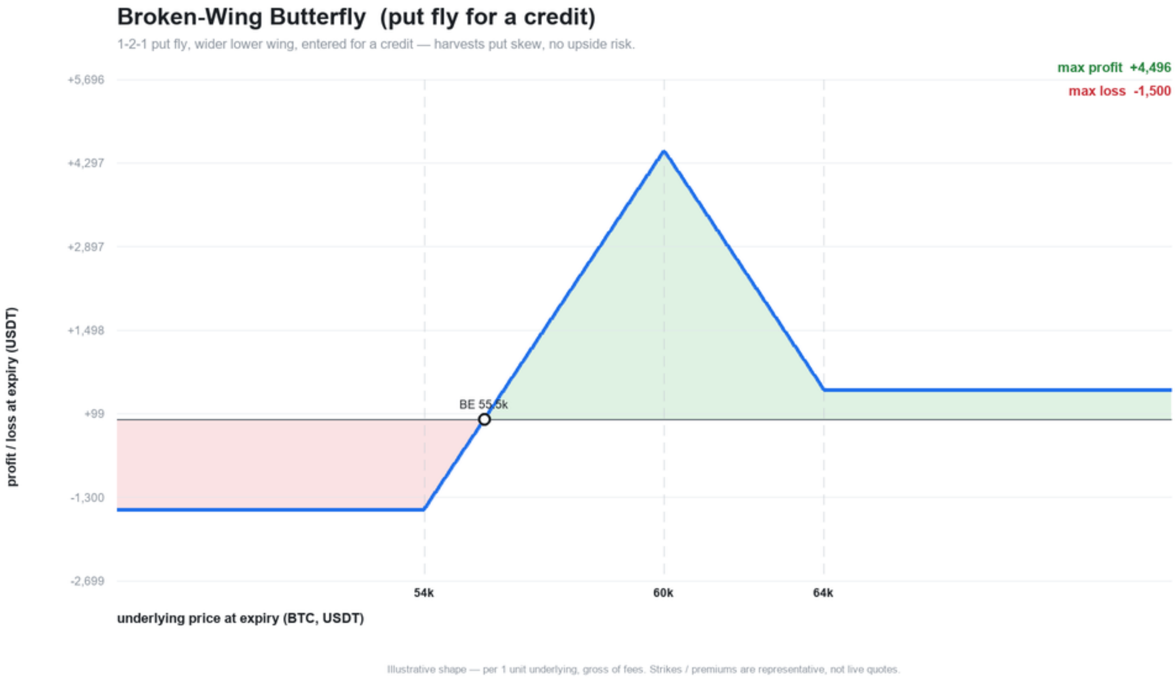
A short put plus a short call spread, sized so the **total credit is at least the call-spread width** — which removes upside risk entirely (the payoff stays positive no matter how far the underlying rallies). Only the downside put tail remains. Picked to harvest call-side premium without taking upside risk in a put-rich regime.



Jade lizard payoff

Broken-wing butterfly (put fly for a credit)

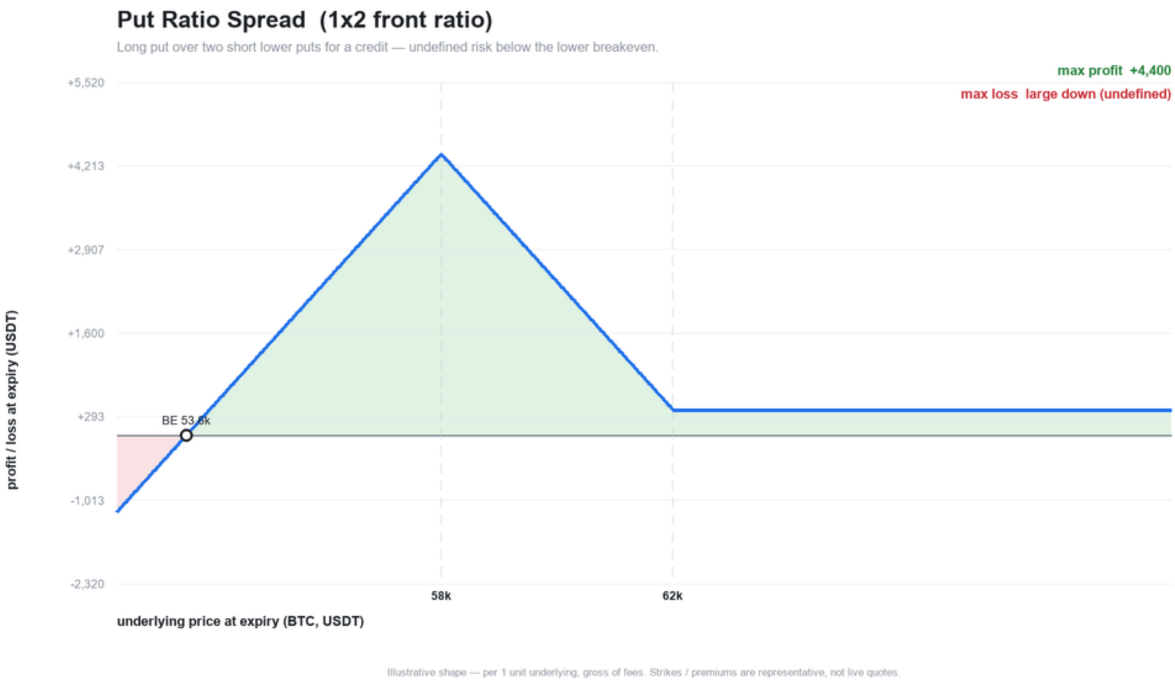
A 1-2-1 put butterfly with a wider lower wing, entered for a credit. Defined risk, no upside risk, and it monetizes put skew — the peak sits where the body strikes are. Picked to harvest put skew with a bounded, credit-financed structure.



Broken-wing butterfly payoff

Put ratio spread

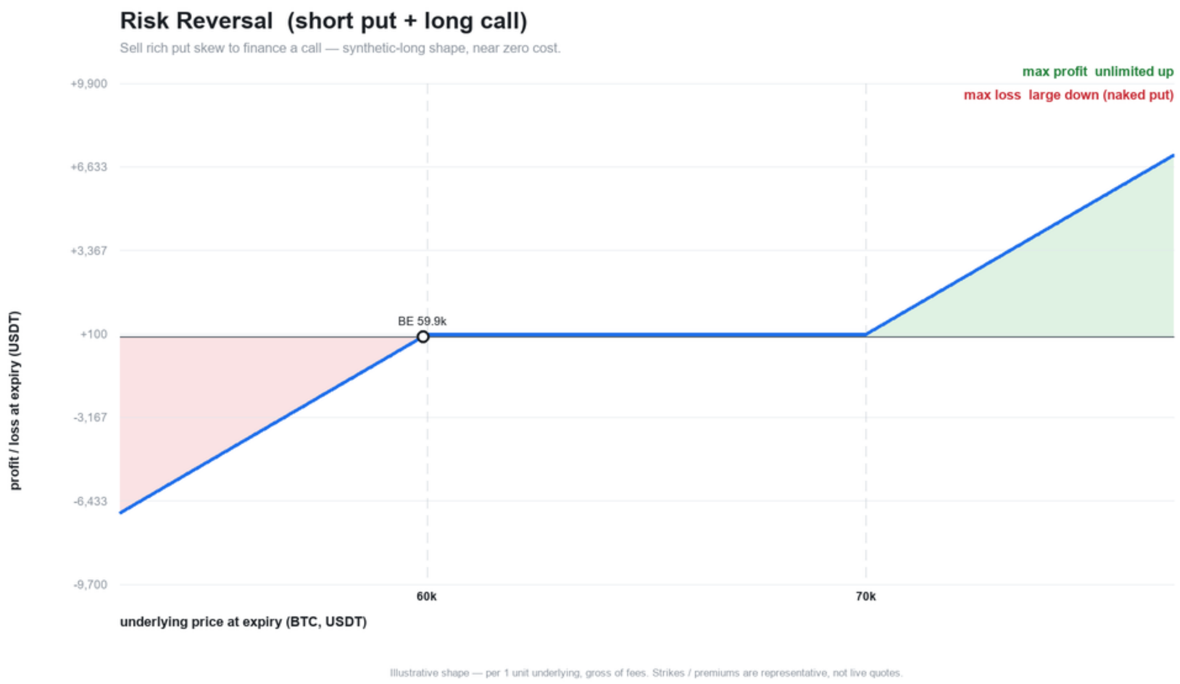
A long put over two short lower puts, entered for a credit. Profits in a mild drift down to the short strikes, but the extra short put leaves **undefined risk below the lower breakeven** — flagged accordingly. Picked for a measured-pullback thesis in a put-skew-rich regime.



Put ratio spread payoff

Risk reversal

Sell a rich OTM put to finance a long OTM call — a synthetic-long shape entered at or near zero cost, monetizing put skew. Uncapped upside, naked put downside. Gated hard against a downtrend and only emitted when skew is genuinely extreme.



Risk reversal payoff

Defensive structures

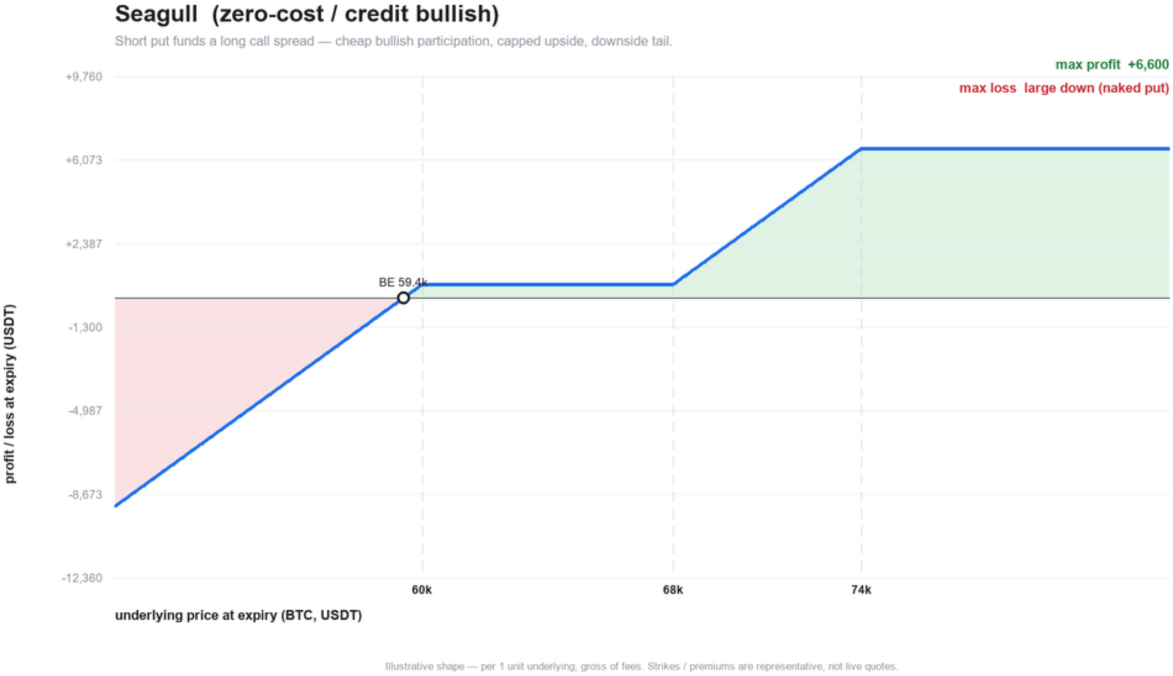
Put ladder

A short put plus **two** lower long puts, entered for a credit. There is a defined pocket of loss between the strikes — but far enough down, a **crash turns the position profitable**. The put structure *for* nervous tape; its downtrend scoring penalty is deliberately mild.

Put ladder payoff

Seagull

A short put funds a long call spread — cheap (zero-cost or credit) bullish participation with a capped upside and a downside put tail. Picked for financed directional exposure when skew lets the put pay for the calls.



Seagull payoff

Long volatility

Calendar

Long a far-dated option and short a near-dated one at the same strike — the library's only **long-vega, debit** structure. It profits if the underlying sits near the strike into the front expiry and benefits from cheap vol and contango, so its scoring **inverts** the VRP / IV-rank / term-structure components: **low-IV regimes surface a calendar instead of "Wait."** (Shown here as value at the front expiry; it is cross-expiry, so it is excluded from the piecewise-linear payoff engine.)

Calendar payoff

04 — Lessons & gotchas

The bugs, surprises, and rules that came out of actually building and running the system.

These are the parts most worth carrying to any similar project.

1. Market-data traps

- **Mark is not a price.** An exchange's option *mark* is a theoretical mid. Position value, profit, and "can I get out?" must be computed from the executable **bid/ask**, not the mark.

This is the single most impactful correction in the project.

- **Slippage-versus-mark explodes on cheap legs.** With a near-zero mark, any "% versus mark" ratio blows up to thousands of percent and paints trivially-cheap buybacks as disasters.

Measure crossing cost as the **half-spread versus the bid-ask mid**, and flag one-sided books explicitly instead of emitting a nonsense number.

- **Average-over-the-smile is not ATM.** Averaging IV over all contracts is dominated by the wings and runs ~8 vol points hot. Comparing that to an external ATM reference manufactures a phantom edge. Always compare **ATM-to-ATM**.
- **Use the perp index for spot.** The option ticker's underlying-price field is unreliable; the linear perpetual's index price is the trustworthy spot.
- **The exchange ignores the base-coin filter on linear tickers.** A "give me BTC contracts" query returns *every* linear contract (other coins, even novelty futures). Filter by symbol prefix.
- **Unified account reports empty per-position option margin.** Real IM/MM come only from the account-level wallet balance; per-leg margin must be reconstructed from the formula.
- `positionValue` **is negative for shorts.** Take absolute value.

2. Money-math traps

- **Fees are ~13% of gross on a premium book.** Model them or your P&L is fiction. Both the open and the close fill are charged; a round trip is two trading fees.
- **The 7%-of-premium fee cap binds on cheap OTM legs** — exactly the legs a strangle-seller trades most. Omitting the cap overstates their cost.
- **Delivery fees exist, both sides, on exercise — except daily options.** OTM expiry is free. The "which term binds" `min(rate × index, 12.5% × intrinsic)` structure is why fee-adjusted breakeven needs a root-finder, not a formula.
- **Contract size is not universal.** BTC/ETH are 0.01 of the underlying per contract; SOL is 1.0. Storing quantity in *underlying units* keeps margin math size-agnostic and prevents 100x errors — but contract *counts* still need the per-underlying size.
- **Never let `inf`/`nan` reach JSON.** An infinite profit factor (no losses yet) must be `null`; infinity breaks the browser's JSON parse and silently blanks the screen.

- **A mid-based credit is not a conservative estimate, it is a different number.** On a put quoted 5 / 25 the mid is 15 and the bid is 5 — **two thirds of the reported credit was spread**. The fix that mattered was not applying a haircut but reporting the decomposition as an **identity** (`mid - net = spread + fees`), because an identity can be tested and a haircut cannot.
- **Publish the headline number your other numbers derive from.** The first version of executable pricing reported the *pre*-fee credit as the headline while deriving max-loss from the *post*-fee one. Every figure was individually defensible and the candidate did not reconcile with itself. A test comparing max-loss against net credit caught it.
- **Two copies of a pricing rule is one copy plus a future bug.** The screener and the roll analyzer each carried their own bid/ask/fee helpers — which is exactly how a fix to one silently misses the other. They now share one module.

3. Modeling discipline

- **Fixed, visible weights can't overfit.** The RV forecast blend uses hand-set weights on purpose. A fitted model would score better in-sample and lie out-of-sample.
- **Sign conventions must be defined once and honored everywhere.** A skew-sign bug (treating negative as put-rich when positive means put-rich) mislabeled and mis-scored every fearful market — live and invisible until audited. Document the sign at every consumption site.
- **Dead-bands beat exact comparisons on noisy signals.** A zero-tolerance moving-average crossover flips its label on microscopic noise; a small dead-band fixes it. A test caught this.
- **Walk-forward or it's an oracle.** The backtest's "world as of day *t*" must exclude every future bar, including in the percentile windows. The slice boundary is the invariant to test.
- **Withhold statistics you can't support.** Percentiles, seasonality marks, and personal edge are all gated on a minimum sample and say "only N so far" below it. Honesty about sample size is a feature, not a caveat.
- **Let calibration tilt, never override.** Personal-edge nudges are expectancy-gated and hard-capped so the market signal and risk limits always dominate. A learning loop that can overpower its guardrails is a liability.
- **Check whether your "signal" is an identity in disguise.** Comparing a leg's theta/gamma breakeven move against the *implied* move looks like a genuine efficiency read. It is algebra: Black–Scholes fixes $\theta = -\frac{1}{2}\sigma^2 S^2 \Gamma$, so the ratio is pinned at ≈0.96 across every strike and tenor. The tell was that it *never varied* — and the fix was to compare against **realized** movement instead. A constant is not a signal; but a constant you can *predict*

makes an excellent data-quality check, which is what that ratio became.

- **Normalize before you compare, or the comparison encodes the tenor.** Roll safety measured as %OTM is not comparable across expiries; measured in expected moves (σ√T) it is. This has a consequence that reads like a bug and is not: a same-strike roll scores *worse* on safety the further out you go, because it buys time while the move budget grows with √T.
- **Score the thing you are about to own, not the thing you are leaving.** A roll's VRP axis prices the *new* contract. Rolling into cheap vol to repair an old position is the classic losing roll, and it is invisible on every other column.
- **Gates must sit beside the score, never inside it.** A weighted average will happily let a strong carry number bury a structural objection like "this is a debit" or "you are selling below forecast RV". Kept separate, a high score with an open gate still reads as a warning.
- **Some warnings should be structural, not predictive.** Roll-chain flags — rolled ≥3×, size growing, cumulative P&L negative — describe the *shape of what already happened*. They make no forecast, so they cannot be wrong, and on an uncapped-tail book that shape is the failure mode itself.
- **Capture the context at the moment, or lose it forever.** "Was I in profit when I rolled?" cannot be reconstructed from a closed ledger. Same principle as the entry-context flywheel: the cheapest data to record is the data that is impossible to backfill.
- **Classify ambiguous samples against your own hypothesis.** A roll chain containing both an opportunistic and a defensive roll counts as **defensive**. The other choice would have quietly biased the evidence in rolling's favour — which is the question being tested.
- **Check that your metric answers the question you think you asked.** A "max sustainable withdrawal" whose only test is *don't hit the ruin floor* approved **2,966/month on a book averaging +555**, while the median account fell from 100k to 38k. Every step was correct; the *question* was wrong. The fix was a second limit — the largest withdrawal leaving the median path with its capital intact — not a better search.
- **Round a safety limit DOWN, never to nearest.** The value was chosen precisely because it clears a threshold; rounding it up can push the number you display back across the very threshold that selected it.
- **Validate the bracket before you bisect.** Doubling the upper bound until it actually *fails* is what makes the search meaningful — bisecting inside a bracket whose top still passes converges to the bracket edge and reports a limit nothing ever tested.
- **State a method's blind spot every time you show its output, not once in a docstring.** A bootstrap cannot produce a month worse than the worst one observed, so on a sample with no crash in it the ruin rate is a *floor*, not an estimate. That sentence ships with the number.
- **When a method has a blind spot, put the compensating measure BESIDE it, not inside it.** The tail budget reprices the current book under a shock, so it does not need the sample to

have contained one. Folding a synthetic tail into the resampling would have hidden which part of the answer came from evidence and which from assumption.

- **Money still owed is not money earned.** Premium collected on open legs is an obligation that can be handed back in full and then some. It is reported next to booked P&L and never added to it — a trader "hitting target" on open premium has hit nothing.
- **Check the target against capacity before checking the pace.** If a monthly target exceeds what the measured distribution supports, it is not a pace problem and no month can fix it. Ordering the checks wrongly would let a structural objection be reported as a schedule.
- **Express pressure as a multiple, not as a difference.** "Needs 3.4x your usual daily rate with 9 days left" changes behaviour; "behind by 400" does not. The risk on an income book is not the shortfall, it is the trade taken *for the target* rather than for the setup.
- **Score a forecast on the information it would have had.** Pin evidence is segmented by the gamma zone and approach direction at *first* capture, never at settlement. Segmenting on anything learned later is lookahead in miniature — the same sin as the backtest slice, at a smaller scale and much easier to miss.
- **Make "not enough evidence" a return value, not a fallback.** The pin-prior lookup returns *insufficient* rather than the pooled number when nothing clears its bar, so a caller cannot quote thin evidence as though it were measured. A belief should resolve to a number or to an admission — never to a default.
- **Raise the sample bar when you cut the sample.** Segmenting splits the same evidence into thinner cells; keeping the pooled minimum while the cells shrink is how a two-observation "rate" gets published.

4. Empirical findings worth remembering

- **The ATM vol-risk premium is razor-thin.** Over ~30-day horizons, implied ≈ realized-forward. A naked-strangle book's edge is therefore **not** ATM richness — it is the OTM **skew/tail** it sells, plus theta/gamma management. This reframed where the book thinks its edge comes from.
- **"Sell when IV rank is high" is validated** (a couple of vol points of edge), while the mid tier is noisy from overlapping-window clustering — and that noise is reported, not hidden.
- **Crypto tails run ~2.7x Gaussian at monthly horizons.** A 2-sigma/30-day put finishes ITM ~6% of the time versus ~2% under normal theory — a direct argument against wide-but-not-wide-enough long-dated naked puts.

5. Product & UX lessons

- **Color the decision, not the number.** When a value's meaning depends on interpretation, make the color track the *decision* (is this a good exit?) and demote the raw detail (spread width) to a tag. Coloring by the wrong dimension made losing-but-tight legs look green.

- **Land on the operational surface.** The first screen should be the live book, because the first question is always "what do I hold and what needs attention?"
- **Group and collapse once it scales.** Flat lists that were fine at five legs become noise at forty. Group per asset with subtotals; hide dense detail behind a toggle whose state survives refresh.
- **Reconcile on every mutation, not just on a timer.** Running the exchange-diff inside the same reload path as every action means an alert clears the instant you act on it, instead of nagging until the next tick.
- **Explicit apply beats debounced auto-run for expensive, deliberate actions.** The scenario stress test dials in three inputs and then runs on click — mid-drag auto-runs were annoying and wasteful.
- **Remove redundant alerting.** A separate notification badge was built and then deleted once the trade brief already surfaced the same priority items. Duplicated nudges erode trust.

6. Engineering & operational lessons

- **Pure core, I/O shell.** Keeping all math in pure functions is what made every model testable offline against synthetic fixtures. It is the structural decision everything else leaned on.
- **Snapshot the account directory before the first await.** Otherwise an account switch mid-request can read one account and write another. This race is silent and data-corrupting.
- **Soft-fail every network dependency.** A dead upstream returns a labeled "unavailable," not an exception, so one outage never blanks a whole screen.
- **Verify by observing the real flow.** Unit tests can't catch a correct formula wired to the wrong column. Drive the UI, read the DOM back, and compare against what the number *should* mean. The most important bugs were found this way.
- `.gitignore` **has no inline comments.** A `pattern # comment` line treats the whole thing as the pattern and silently ignores nothing — which once let plaintext credentials get tracked. Comments go on their own line, and the ignore rules are the real secret-leak guard.
- `git add -A` **does not un-track already-tracked files** even after you ignore them; they must be explicitly removed from the index. Always scan the staged diff for secrets before a commit that could ever go public.
- **Encryption at rest is defense-in-depth, not the primary control.** The primary control is never committing the secret in the first place.

7. The one-line summary

Build the boring, honest plumbing first — executable prices, real fees, correct margin, lookahead-free backtests, sample-gated statistics — and *then* the synthesis on top is trustworthy. A pretty score on top of a fictional mark is worse than no score at all.

07 — Mathematical foundations & references

Every formula the system computes, stated precisely, with its provenance.

This document exists because a decision-support tool is only as trustworthy as the weakest formula inside it. Where a model comes from the literature, it is cited. Where it is an exchange's published mechanic, the exchange is the authority. Where it is a practitioner construct with no canonical definition, that is **said out loud** rather than dressed up as theory. Where it is this system's own choice, the choice and its cost are stated.

Provenance tags appear on every model:

Tag	Meaning
[L]	Peer-reviewed literature — the formula is standard and the citation is canonical
[X]	Exchange-documented mechanic — the venue's own published rule, verified against it
[P]	Practitioner construct — widely used, no canonical academic definition
[S]	This system's own choice — a deliberate modelling decision, not a citation
[M]	This system's own measurement — an empirical finding from its own data

Numbered references are collected in §12.

1. Notation and conventions

Symbol	Meaning
S	spot / index price of the underlying
F	forward price for a given expiry
K	strike
T	time to expiry, in years
σ	volatility, annualized, as a decimal (0.55, not 55)
r	risk-free rate — **taken as 0 throughout** (see §3.3)
Δ,Γ,ν,Θ	the greeks; ν per **vol point**, Θ per **day**
O,H,L,C	daily open, high, low, close
n	sample size (bars, trades, months — always stated per formula)

Conventions fixed once, applied everywhere:

- **Annualization uses √(365), not √(252). [S]** Crypto trades every calendar day. Using the equity convention would inflate every volatility figure by √(365/252) ≈ 1.20 — a 20% error that would look entirely plausible.
- **Quantity is stored in underlying units** (e.g. BTC), never in contract count. Contract count is derived as q / contract size, and contract size is a **venue** fact, not a global one (§8.4).
- **Implied volatility is stored as a decimal.** Venues differ — one publishes `0.5521`,

another publishes `55.21` vol points. Normalizing at the adapter boundary is mandatory; a factor of 100 in σ is not subtle in a payoff but is nearly invisible in a rank.

- **Skew sign:** positive = puts richer (§6). Defined once, restated at every consumption site, because a sign flip here is silent and directional.

2. Realized-volatility estimators

All four estimators below produce a **daily variance** σ^2_d which is annualized as $\sigma = \sqrt{365\ \sigma^2_d}$. All operate on completed bars only — the in-progress candle is always dropped, since a partial bar's high/low understates the day's true range and biases every range-based estimator downward.

2.1 Close-to-close [L]

The textbook baseline, with $r_i = \ln(C_i / C_{i-1})$:

$$\sigma^2_{cc} = (1)/(n-1)\Sigma_{i=1}^{n}(r_i - r)^2$$

Kept only as a reference point. It discards the intraday range entirely and is the noisiest of the four.

2.2 Parkinson (1980) [L] — ref. [1]

$$\sigma^2_P = (1)/(4n\ln 2)\Sigma_{i=1}^{n}[\ln((H_i)/(L_i))]^2$$

Uses the high–low range. Under driftless geometric Brownian motion its variance is roughly **one fifth** that of the close-to-close estimator — the efficiency gain that motivates the whole range-estimator family. It has a known downward bias in discretely-sampled data, because the observed high and low understate the continuous extremes.

2.3 Garman–Klass (1980) [L] — ref. [2]

$$\sigma^2_{GK} = (1)/(n)\Sigma_{i=1}^{n}[½\ln^2((H_i)/(L_i)) - (2\ln 2 - 1)\ln^2((C_i)/(O_i))]$$

Adds open/close information to Parkinson and is more efficient again. Both Parkinson and Garman–Klass **assume zero drift**, which is why neither is the primary estimator here.

2.4 Rogers–Satchell (1991) [L] — ref. [3]

$$\sigma^2_{RS} = (1)/(n)\Sigma_{i=1}^{n}[\ln(H_i)/(O_i)\ln(H_i)/(C_i) + \ln(L_i)/(O_i)\ln(L_i)/(C_i)]$$

Drift-independent — this is the property that matters for an asset that can trend 40% in a month. It appears here as a component of Yang–Zhang rather than on its own.

2.5 Yang–Zhang (2000) [L] — ref. [4] — *the primary estimator*

Decomposes total variance into overnight, open-to-close, and intraday-drift-free parts:

$$\sigma^2_{YZ} = \sigma^2_o + k\,\sigma^2_c + (1-k)\,\sigma^2_{RS}$$

with

$$\sigma^2_o = Var[ln(O_t)/(C_{t-1})],\ \sigma^2_c = Var[ln(C_t)/(O_t)],\ k = (0.34)/(1.34 + (n+1)/(n-1))$$

Both variances are sample variances with the n-1 denominator. The weight k is Yang and Zhang's own minimum-variance choice and is **not tuned here**.

The crypto caveat, stated because it changes what the estimator is. Yang–Zhang's advantage over Rogers–Satchell comes from separating the *overnight gap* — the risk that accrues while the market is closed. Crypto perpetuals never close, so consecutive candles are contiguous and $\sigma^2_o \approx 0$. On this data YZ therefore degrades gracefully into a k-weighted blend of open-to-close and Rogers–Satchell. It remains the best of the four, but the specific efficiency figures quoted in the original paper are derived under an *equity* open/close structure and should not be claimed verbatim here. **[M]**

3. Option pricing and the greeks

3.1 Black–Scholes–Merton [L] — refs. [5], [6]

Used for delta reconstruction where a venue publishes a coarse strike grid, for scenario repricing of the book under spot/IV/time shocks, and for probability-of-profit estimates. With r = 0:

$$d_1 = (ln(S/K) + ½\sigma^2\ T)/(\sigma\sqrt{T}),\ d_2 = d_1 - \sigma\sqrt{T}$$

$$C = S\ N(d_1) - K\ N(d_2),\ P = K\ N(-d_2) - S\ N(-d_1)$$

3.2 Black-76 on the forward [L] — ref. [7] — *the cross-venue greeks*

Options on **forwards** (which is what a dated crypto option effectively is, since each expiry has its own forward) are priced by Black's 1976 model. With r = 0:

$$d_1 = (ln(F/K) + ½\sigma^2\ T)/(\sigma\sqrt{T})$$

$$\Delta_{call} = N(d_1),\ \Delta_{put} = N(d_1) - 1$$

$$\Gamma = (\varphi(d_1))/(F\ \sigma\sqrt{T}),\ v = (F\ \varphi(d_1)\sqrt{T})/(100),\ \Theta = -(F\ \varphi(d_1)\ \sigma)/(2\sqrt{T} \cdot 365)$$

where φ is the standard normal density. The /100 on vega expresses it **per vol point**, and the /365 on theta expresses it** per day** — both are display conventions, chosen to match what the venues themselves publish so the two are interchangeable.

Why this specific model matters here. The system must produce greeks for a venue that publishes some but not all of them, and those greeks must be *comparable* to the other venue's. Black-76 on the per-expiry forward reproduces the venue's own published greeks to **worst error 0.00002 in delta and 0.03% in vega** across the live chain. **[M]** That agreement is the evidence that the model, the forward, and the IV units were all read correctly — three things that would each be silently wrong on their own.

3.3 Why r = 0 [S]

There is no risk-free rate in a crypto options book in the sense Black–Scholes means. Setting r=0 and pricing on the **forward** puts all the carry — funding, basis, and the cost of the coin — into F, where the venue has already measured it and publishes it per expiry. Introducing a separate r would double-count it.

A related trap, verified live: both venues publish a per-expiry **forward** (`underlyingPrice` / `underlying_price`) *and* an **index**. They are not the same number — on one observation the forward sat **3.93% above** the index. **[M]** Greeks are computed on the forward; money conversions use the index. Using one where the other belongs is a ~4% error in the moneyness of every strike.

3.4 Inverse (coin-margined) options — the self-quanto payoff [L]/[X] — ref. [8]

A coin-margined option quotes its premium *in the underlying* and settles *in the underlying*. Its payoff per contract is therefore

$$Payoff_{put}(S_T) = (max(K - S_T, 0))/(S_T), Payoff_{call}(S_T) = (max(S_T - K, 0))/(S_T)$$

denominated in the coin. This is a **self-quanto** structure — the payoff is divided by the very price it is written on — and it is *not* a linear payoff even though the underlying option is. The short-put P&L in coin is

$$P\&L = q(premium_{coin} - (max(K-S_T, 0))/(S_T))$$

Three consequences the system depends on:

1. **The curve bends.** The coin loss on a short put grows more slowly than the USD loss, because the denominator grows as price falls — no, it *shrinks*, which makes the coin loss grow **faster** than linearly in the region that matters. The distinction is not academic: it changes the shape of the drawdown.
2. **The breakeven moves.** Solving q prem = qmax(K-S,0)/S gives S^* = K/(1 + prem), not K - prem. On a real book these differed by **$261**. **[M]**
3. **A USD-denominated payoff engine cannot be reused.** Subtracting a USD intrinsic from a coin premium is dimensionally meaningless, and — because the premium is numerically tiny in coin terms — it produces a curve that looks *almost right*. See 06 — Lessons.

4. The variance / volatility risk premium

4.1 Definition as used here [S], grounded in [L] — refs. [9], [10], [11]

$$VRP_{pts} = \sigma_{IV} - \sigma_{RV}^{fcast}, VRP_{ratio} = (\sigma_{IV})/(\sigma_{RV}^{fcast})$$

The academic construct (Carr–Wu [9], Bollerslev–Tauchen–Zhou [11]) is defined in

variance terms and against *realized-forward* variance — i.e. what volatility actually turned out to be over the option's life. This system reports it in **volatility points** against a **forecast**, because it is used as a live decision gate before the outcome exists. The two are related but not identical, and the difference is stated wherever the number is displayed.

Bakshi–Kapadia [10] is the cleanest evidence that the premium is real and negative for the buyer: delta-hedged option positions lose money on average, which is the seller's edge stated from the other side.

4.2 The RV forecast: HAR-inspired, deliberately unfitted [S], after [L] ref. [12]

$$\sigma^{fcast} = 0.30\ \sigma_{YZ}^{(5)} + 0.40\ \sigma_{YZ}^{(20)} + 0.30\ \sigma_{YZ}^{(60)}$$

Corsi's HAR-RV [12] regresses realized variance on daily / weekly / monthly components with **OLS-fitted** coefficients, exploiting the long-memory structure of volatility. This system keeps the multi-horizon *structure* and throws away the *fitting*:

- **Windows are 5 / 20 / 60 days**, not Corsi's 1 / 5 / 22.
- **Weights are fixed at 0.30 / 0.40 / 0.30** and are never estimated from data.
- When a window has insufficient history it is **dropped and the remaining weights renormalized**, so a young dataset yields a coarser but honest forecast rather than none.

This is a real accuracy sacrifice, taken on purpose: a fitted forecast on a single underlying's short history would overfit, and — more importantly — could not be audited by the person trusting it. Every weight here is visible in one line.

4.3 The headline empirical finding [M]

Measured on this book's own BTC history: the **ATM** vol-risk premium is razor-thin — implied ≈ realized-forward over ~30 days. The edge in this book is therefore **not** ATM richness; it is the **OTM skew and tail being sold**, plus theta/gamma management. This is a finding that argues against the strategy's own marketing, which is why it is recorded.

5. Expected move, and the √(2/π) correction

5.1 The straddle-implied move [L] — ref. [13]

$$EM\% = (P_{ATM} + C_{ATM})/(S)$$

using the nearest-to-spot strike with **both sides quoted**.

The precision point most tools get wrong. Brenner and Subrahmanyam [13] show that at the money, with r=0,

$$P_{ATM} + C_{ATM} \approx \sqrt{((2)/(\pi))}\ S\ \sigma\sqrt{(T)} \approx 0.7979\ S\ \sigma\sqrt{T}$$

So the straddle price is **not** a one-sigma move. It is ≈ 0.80σ — which is exactly E[|X|] for X □ $N(0,\sigma^2)$, the **mean absolute deviation**. Calling the straddle "the expected move" is therefore correct in the literal sense (it *is* the expected absolute move) and wrong if read as a one-standard-deviation band, which contains ~68% of outcomes rather than the ~58% this does.

The system compares EM% against the **realized** distribution of |rm move| over the same horizon, which sidesteps the confusion entirely: mean-absolute is compared with mean-absolute.

5.2 The overlapping-window caveat [L] — ref. [14]

The realized-move distribution is computed over **rolling, overlapping** h-day windows. This is the standard trade-off for sample size on a short history, and it is standard that overlapping windows induce serial correlation and **inflate the effective n** — the Hansen–Hodrick problem [14]. The system reports the raw n and does not compute confidence intervals from it, because an interval built on an inflated n would be worse than no interval.

6. Skew

6.1 The 25-delta convention [P]/[L] — ref. [15]

$$Skew_{25} = \sigma_{IV}(\Delta_{put} = -0.25) - \sigma_{IV}(\Delta_{call} = +0.25)$$

Positive = puts richer, the normal fearful state. This is the FX **risk-reversal** convention (Clark [15]) applied to crypto, where the same 25Δ market convention has been adopted by the major venues.

Two disciplines around it:

- **The sign is defined once and restated at every consumption site** (labelling, digest, scoring). A sign error here inverts the reading of every fearful market and is completely invisible in the output — this exact bug shipped and ran live before an audit caught it.
- **Delta is reconstructed via Black-Scholes, not trusted from the strike grid**, and the value is **withheld** when the nearest available strike is not genuinely near 25Δ. A coarse grid must not be permitted to produce a precise-looking number.

6.2 What is *not* claimed

The system does **not** compute model-free risk-neutral skewness (Bakshi–Kapadia–Madan [16]), which would require a full strike continuum and a different integral. The 25Δ measure is a two-point proxy, and is labelled as such.

7. Gamma positioning

This section carries the heaviest **[P]** weighting in the document, and that is exactly why the system *measures* its predictions rather than assuming them (§7.4).

7.1 Max pain [P], with academic cousin [L] — refs. [17], [18], [19]

$$K^* = argmin_K\ \Sigma_i OI_i \cdot intrinsic_i(K)$$

i.e. the strike minimizing total option-holder payout across the expiry.

"Max pain" itself is a practitioner heuristic with no academic derivation. What the literature establishes is narrower and better evidenced: prices *cluster at strikes* on expiration dates (Ni, Pearson & Poteshman [17]), and there is a mechanism — delta-hedging rebalancing by writers — that produces it (Avellaneda & Lipkin [18]); Golez & Jackwerth [19] find pinning in S&P 500 futures. Related but not the same claim. The honest statement is: *strike pinning is documented; "price drifts to the single max-pain strike" is folklore until measured.*

7.2 Gamma exposure (GEX) [P]

$$GEX(K) = \Gamma_{call}(K)\ OI_{call}(K)\ S^2 - \Gamma_{put}(K)\ OI_{put}(K)\ S^2$$

(scaled by a constant for display). The sign convention assumes dealers are **long calls, short puts** relative to the retail flow — the standard practitioner assumption, and one that is *an assumption*, not a measurement. Positive net GEX is read as a dealer long-gamma zone (hedging is mean-reverting: buy dips, sell rallies), negative as short-gamma (hedging amplifies: sell dips, buy rallies).

The underlying mechanism — that option hedging feeds back into the underlying's dynamics — *is* documented (Ni, Pearson, Poteshman & White [20]; Barbon & Buraschi [21]). The specific GEX construction is not.

7.3 Zero-gamma flip [P]

The spot level where the cumulative net-GEX profile crosses zero: the boundary between the stabilizing and accelerating regimes. Used as a summary field, a digest signal, and a scoring input.

7.4 Why the system stores pin history [S]/[M]

Because §7.1–7.3 are practitioner constructs, the system does not take them on faith. It snapshots, per (underlying, expiry) per day, the max-pain level, the gap to spot, the net GEX, and the gamma zone at spot. Once an expiry passes, the row sequence becomes a **completed sample**: did $|S - K^*|$ actually narrow between the first and last pre-expiry capture, and did the gamma zone at capture time predict it?

Three honesty rules make that sample meaningful:

- An expiry needs **≥2 captures** to measure convergence at all; a single row is excluded.
- The report needs **≥5 completed expiries** before quoting any rate, and always carries n.
- **Daily expiries are segmented out.** The pinning literature concerns *serial* expirations carrying real open interest. Crypto dailies carry very little, so pooling them with the

monthlies would drown the signal in noise that looks like data. Segmenting raises the minimum-n bar rather than keeping it fixed while the cells get thinner.

7.5 The two pin tests, and the evidence hierarchy [S]

Two different statistics are computed, and they are **not** interchangeable.

Settlement rate — the real test. Over recorded settlements, the fraction that printed closer to the pin than they started, and the fraction that settled within a tolerance of it:

$$moved\ toward = (\#\{\,|S_{settle} - K^{*}| < |S_{first} - K^{*}|\,\})/(n_{settled})$$

Convergence rate — a proxy. Over completed expiries, whether the gap narrowed between the first and last pre-expiry capture, with $g_t = |S_t - K^{*}_t| / S_t$:

$$converged\ iff\ g_{last} < g_{first}$$

Convergence is weaker evidence: a gap can narrow and the expiry still settle far away. It exists because it accumulates faster — every completed expiry contributes, whereas settlement evidence needs a recorded settle price.

Segmentation is on the state at FIRST capture — approach direction ($sgn(K^{*} - S)$: is the pin above or below?) and the gamma zone at spot — because that is the information a forecast would actually have had. Segmenting on anything learned later is a lookahead in miniature.

The evidence hierarchy, applied when the advice layer asks for a pin prior:

$$settlement\ succ\ convergence,\ direction\ succ\ zone\ succ\ pooled$$

— the most trustworthy source first, and within it the most *specific* segment that clears its own minimum, falling back to pooled. When **nothing** clears, the function returns `sufficient: false` with **no rate at all**, and callers are required to report the pin as *unmeasured*. That fallback is the entire purpose of the store: it forces a practitioner belief to resolve into either a number or an explicit admission, never into a default assumption that pins hold.

8. Fees, margin, and breakeven

8.1 Trading fee [X]

$$fee_{fill} = min(rate \times S_{index},\ c_{prem} \times premium) \times q$$

with taker rate ≈ 0.03%, maker ≈ 0.02%, and the **premium cap** c_{prem} being a **venue** parameter: **7%** on one venue, **12.5%** on the other. The cap is the subtle term — on far-OTM cheap legs the percentage-of-premium term is smaller than the percentage-of-index term, so the cap binds precisely on the legs a strangle seller trades most. Ignoring it overstates the fee where it matters.

A round trip pays this on **both** the open and the close fill. Modelling one side halves the cost.

8.2 Delivery (settlement) fee [X]

$$fee_{delivery} = min(deliveryRate \times S_{index}, 12.5\% \times intrinsic) \times q$$

charged **only on exercise**. An out-of-the-money expiry is free. `deliveryRate` ≈ 0.015% for BTC/ETH, ≈ 0.02% for alts. **Daily options incur no delivery fee at all** — an exception re-verified against the exchange's own help page, and one that materially changes breakeven math (§8.5).

8.3 Initial margin, short option [X]

$$IM = [max(f_{max}S - OTM, f_{min}S) + max(avg\_price, mark)] \times q$$

$$OTM_{call} = max(0, K - S), OTM_{put} = max(0, S - K)$$

with (f_{max}, f_{min}) ≈ (0.10, 0.05) for BTC/ETH, verified against the exchange's own worked example. A long option's margin is its premium.

This formula describes exactly one venue. It is reconstructed by hand because the unified account reports empty per-position IM for options — only an account-level total — and per-leg attribution is what tells the trader *which* leg to trim. On a venue that publishes no IM factors of its own, the system now **refuses to present this number as modelled** rather than computing a confident figure with another exchange's parameters.

8.4 Contract size is a venue fact [X]

$$contracts = (q)/(contract\ size(venue, underlying))$$

0.01 BTC on one venue, **1 BTC** on another; **1.0** for SOL where BTC/ETH are 0.01. Because q is stored in underlying units, the margin and payoff formulas are size-agnostic — only contract *counts* need it. This single convention prevents a family of 100× errors, and every place it was violated produced exactly that: a 0.9-contract book reported as 90.

8.5 Fee-adjusted breakeven requires a root-finder [S]

The terminal price at which a leg nets exactly zero **after** the open trading fee **and** the delivery fee has no closed form, because the delivery fee is a min(·,·) whose binding term depends on the premium size. So the breakeven is obtained by **bisection against the exact fee model** [22] rather than a formula:

- the net-at-expiry P&L is monotone on each side of the strike, so the root is bracketed;
- bisecting against the *same* fee function used on real closes guarantees the displayed breakeven and the realized P&L agree to the cent — same code path, different caller;
- for daily options the delivery term drops and the breakeven collapses to the clean $K - (entry - fee_{open})$ for a short put.

A short put's fee-adjusted breakeven shifts **up** toward the strike (less room); a short call's shifts **down**; longs need a bigger move to overcome the drag.

8.6 Executable pricing: the decomposition and its identity [S]

For a structure of legs ℓ with sides $s_\ell \in \{+1 \text{ (sell)}, -1 \text{ (buy)}\}$ and quantities q_ℓ, define the executable price of each leg as the side of the book the fill actually takes:

$$p_\ell = \begin{cases} bid_\ell & s_\ell = +1 \\[2pt] ask_\ell & s_\ell = -1 \end{cases} \qquad m_\ell = \tfrac{1}{2}(bid_\ell + ask_\ell)$$

Then

$$C_{mid} = \Sigma_\ell\, s_\ell\, m_\ell\, q_\ell, \quad C_{exec} = \Sigma_\ell\, s_\ell\, p_\ell\, q_\ell, \quad C = C_{exec} - \Sigma_\ell\, fee_\ell$$

with fee_ℓ from §8.1. The two costs of reality are named separately:

$$\text{spread cost} = C_{mid} - C_{exec}, \quad \text{entry fees} = \Sigma_\ell\, fee_\ell$$

giving the **auditable identity** that every candidate must satisfy:

$$C_{mid} - C = \text{spread cost} + \text{entry fees}$$

and the retention ratio

$$\eta = \frac{C}{C_{mid}}$$

Why an identity rather than just a pessimistic number. Any monotone haircut would make the screener more conservative. An identity makes it **checkable**: the three reported figures cannot drift apart without the equation failing, which is what a test asserts. The alternative — a single "adjusted credit" — is unfalsifiable by construction.

Per-leg net price. Each leg also carries $p_\ell - fee_\ell/q_\ell$ for a sold leg and $p_\ell + fee_\ell/q_\ell$ for a bought one, so the payoff engine computes breakevens that already include fees **without knowing anything about fee models**. Fees are folded in at the only place that knows about them.

What is deliberately absent. Exit costs. A screener ranks *entries*; what it costs to leave depends on when and why, and is priced by the roll analyzer (§9.6) and the P&L calculator instead. Mixing the two would double-count the spread on positions held to expiry, which is the majority of them on this book.

9. Position management: theta efficiency and rolls

§9.1–9.5 cover the carry on a leg you hold; §9.6–9.7 cover the decision to replace it.

Theta efficiency is the most mathematically load-bearing piece in the system, because the obvious version of it is **circular** and produces a number that looks informative and is not.

9.1 The daily P&L decomposition [L] — ref. [23]

For a delta-hedged short option over one day,

$$\Delta\, P\&L \approx |\Theta|\, \Delta\, t - ½\, \Gamma\, (\Delta\, S)^2$$

so the move that exactly consumes one day of theta is

$$\Delta\, S^* = \sqrt{((2\, |\Theta|)/(\Gamma))}$$

9.2 Why comparing Δ S^* to the *implied* move is circular [L]/[M]

The Black–Scholes PDE with r = q = 0 gives

$$\Theta = -½\, \sigma^2\, S^2\, \Gamma$$

Substituting into Δ S^* collapses it exactly:

$$\Delta\, S^* = \sqrt{((2 \cdot ½\, \sigma^2\, S^2\, \Gamma\, /\, 365)/(\Gamma))} = (S\sigma)/(\sqrt{(365)})$$

— which *is* the implied daily move. The comparison is an identity, not a signal. Verified against the live BTC chain: Δ S^* / implied sits at a flat **≈0.96** across every strike and every tenor. **[M]** The residual 4% is the venue's theta carrying a rates/carry term that the ½Γ(Δ S)^2 approximation drops.

9.3 The comparison that does carry information [L] — refs. [24], [25]

$$edge\ ratio = (\Delta\, S^*)/(realized\ daily\ move)$$

Comparing the breakeven move against **realized** movement is the variance risk premium measured *on this specific leg with its own greeks*, rather than as a surface average. This is the discrete, per-leg form of the classical delta-hedged P&L result (Carr–Madan [24]; El Karoui, Jeanblanc-Picqué & Shreve [25]):

$$P\&L_{hedged} = ½\int_0^T \Gamma_t\, S_t^2 (\sigma^2_{implied} - \sigma^2_{realized,t}) dt$$

Realized above breakeven □ the position **loses in expectation** even while theta prints positive every single day. That is the failure mode the whole module exists to surface. Verdict bands: ≥1.15 *paid well*, ≥1.00 *marginal*, below *underpaid*. **[S]**

9.4 The stale-greeks detector [S]

Because §9.2 establishes that Δ S^*/implied *must* be ≈0.96, any leg whose ratio wanders more than ±0.20 from it has greeks that are stale or internally inconsistent, and its efficiency reading is flagged untrustworthy. A degenerate identity turned into a data-quality check.

9.5 The units trap [S]

On an enriched leg, Θ is **position** theta (already × q, signed so a seller earns positive) while Γ and ν are **per-unit and unsigned**. Mixing the two scales leaves a stray q inside the square root and yields a plausible, wrong breakeven. Everything converts to per-unit first, and the function signature exists to force it.

Two further per-leg readings come from the same greeks. **Vol-spike fragility**, as the number of days of theta that one vol point costs:

$$D_v = (|v|\ q)/(\Theta)$$

and **days to collect** the premium still outstanding, $D_c = \text{mark} \cdot q/\Theta$.

A leg needing many days of carry to earn back a one-point IV rise is fragile regardless of how good its edge ratio looks.

9.6 Roll advantage score [S]

A roll's *price* (§8.6, applied twice — the old leg is bought back at the ask, the new one sold at the bid, and a fee is paid on both fills) answers "what does this pay?" but not "is this a good trade?". Five axes are scored on [0,100] and weight-averaged:

$$Score = (\Sigma_{a \in A}\, w_a\, f_a(x_a))/(\Sigma_{a \in A}\, w_a)$$

where A is the set of axes that could actually be computed — **weights renormalize over A**, so a missing greek feed lowers the score's resolution rather than silently scoring that axis zero. Each f_a is a **piecewise-linear, clamped** map from a raw value to a subscore, with breakpoints fixed in advance and visible.

Axis	w_a	Raw quantity x_a
carry	0.25	credit per day ÷ collateral, annualized (× 365) — the yield on the time bought
vrp	0.25	$\sigma_{IV}^{new} / \sigma_{RV}^{fcast}$ — on the contract being **opened**
safety	0.25	$(\lvert K_{new} - S\rvert)/(S\ \sigma_{new} \sqrt{(T_{new})})$ — strike distance in expected moves
gamma	0.15	$((\lvert\Theta\rvert/\Gamma)_{new})/((\lvert\Theta\rvert/\Gamma)_{old})$ — carry per unit of convexity
capital	0.10	IM_{new} / IM_{old} (§8.3)

Why safety is measured in σ√T and not in %OTM. Raw distance is not comparable across tenors: 8% away is a wall at 2 DTE and nothing at 60. Normalizing by the expected move over the *contract's own remaining life* makes candidates at different expiries directly comparable. The consequence is worth stating because it looks like a bug and is not: a **same-strike roll scores progressively worse on safety as the tenor grows**, because it buys time while the move budget grows as √(T). It is buying time, not safety, and the axis says so.

Why the VRP axis prices the new contract. Rolling into cheap vol to repair an old position is the classic losing roll, and it is invisible unless the leg being *opened* is checked against forecast RV.

Gates are separate from the score. Two conditions — a debit roll, and selling below forecast RV ($x_{vrp} \le 1$) — cap the verdict at *questionable* regardless of the weighted average, and a new strike inside one expected move ($x_{safety} < 1$) or a margin rise above 50% is reported as an open gate. Folding gates into the score would let a strong carry number average away a structural objection; kept separate, a high score with an open gate still reads as a warning.

9.7 Roll-chain evidence and its gates [S]

Legs linked by a roll chain are bucketed by the **state at roll time**, and average P&L is compared:

never rolled | *rolled while in profit* | *rolled while tested*

Four rules make the comparison honest rather than flattering:

1. **Only completed chains are scored.** An open chain's P&L is not decided; including it biases the result in whichever direction the position currently sits.
2. **A chain containing both kinds of roll counts as *tested*.** The defensive roll is the risk-relevant event; classifying a mixed chain as clean would bias the comparison in rolling's favour.
3. **Roll context is captured at roll time**, on the new leg. "Was I in profit when I rolled?" cannot be reconstructed from a closed ledger.
4. **Sample gates rise with the cut**: an overall minimum of completed chains before any verdict, *and* a per-bucket minimum before two buckets are compared. Below either, counts are shown and the verdict is withheld.

Martingale warnings are **structural, not predictive** — they describe the shape of what has already happened rather than forecasting an outcome: rolled ≥ 3 times; size grew across the chain; cumulative realized P&L negative; every roll made while losing. On a book with an uncapped tail, a long roll chain with growing size is the classic failure pattern, and each individual roll looks like "collect more credit" right up until it doesn't.

10. Statistical machinery

10.1 Percentile rank with a sample gate [S]

Today's reading is ranked against the book's own trailing history (typically 90 days) as a true empirical percentile, with a 5-day movement arrow attached. **Below a minimum sample the percentile is withheld or explicitly hedged** — "only N days tracked so far" is a first-class output, not an error state.

10.2 Bootstrap for withdrawal sustainability [L] — refs. [26], [27]

Monthly returns are resampled **with replacement** from the months actually observed (Efron [26]), compounding equity proportionally and taking the withdrawal in cash after each month's return:

$$E_{t+1} = E_t\,(1 + r_{i_t}) - W, \quad i_t \sim \mathrm{Uniform}\{1, \ldots, n\}$$

Ruin is declared below 30% of starting capital. The simulation is **seeded**, so the number does not move between page reloads.

Why bootstrap rather than fit. Fitting a parametric distribution to a dozen observations of a negatively-skewed, fat-tailed return stream produces a clean number worth nothing. Resampling is honest about the evidence — at one specific, stated price:

> ***A bootstrap cannot produce a month worse than the worst month observed.***

If the sample has not lived through a crash, every path is optimistic and the ruin rate is a **floor, not an estimate**. This is the classical limitation of resampling in the tail (Embrechts, Klüppelberg & Mikosch [27] is the standard treatment of what would be required instead). It is why the tail budget is computed *separately*, by repricing the **current book** under a shock — a number that does not depend on the sample having contained one.

Modelling choices, all explicit **[S]**: monthly returns are taken as P&L / capital (so a materially different capital base today makes the estimate wrong, and the report says so); equity compounds proportionally, so a shrinking account earns proportionally smaller premium — holding size constant as equity falls would flatter every path.

10.2a Two different withdrawal questions [S]

Let R(W) be the bootstrapped ruin rate at withdrawal W (paths ending below 30% of starting capital) and M(W) the **median** terminal equity. Two limits can be defined, and they are not the same number:

$$W_{sust} = max\{ W : R(W) \leq varepsilon \}, W_{pres} = max\{ W : M(W) \geq E_0 \}$$

W_{sust} is the obvious construction and **not the one people mean**. Its test only forbids *falling below the ruin floor*; it is entirely satisfied by a path that consumes most of the account on the way. Measured on the reference book:

	value
average month	+555
W_{sust} at a 5% ruin tolerance	**2,966 / month**
median terminal equity under that withdrawal	**100k → 38k**
W_{pres}	far lower

A withdrawal that shrinks the median account by 62% over the horizon is capital depletion on a schedule, not income. **[M]** W_{pres} — the largest withdrawal leaving the median path whole — is what the cadence layer (§10.6) checks a target against.

Both are found by **bisection**, after doubling the upper bracket until it actually fails so the bracket is valid, and both are **rounded down, never to nearest**:

$$W_{reported} = (lfloor 100 W rfloor)/(100)$$

A safety limit is selected *because* it satisfies a threshold; rounding it upward can push the reported figure back across the very threshold that chose it. A returned **0** is a real result, not a missing value: even withdrawing nothing breaches the tolerance.

10.2b Tail budget — the unit that makes an uncapped tail legible [S]

$$\text{months of income} = \frac{|\Delta Pi_{stress}|}{Pi_{month}}$$

where ΔPi_{stress} comes from **repricing the live book** under a spot/IV/time shock (§3.1 pricing, applied to current positions), not from the sample. This is deliberate: §10.2's bootstrap cannot invent a crash, so the stress number must come from somewhere that does not depend on one having happened. Premium selling accumulates income linearly and returns it in jumps; this expresses the size of one jump in the only unit the trader budgets in.

10.3 Sequence-of-returns risk [L] — ref. [28]

The reason a "monthly income" framing needs simulation at all rather than an average: with withdrawals, the **order** of returns changes the outcome, and a negatively-skewed return stream concentrates the damage. Bengen [28] is the canonical treatment for withdrawal rates; the mechanism is identical here even though the asset is not.

10.4 Trade statistics [S]

$$\text{expectancy} = \frac{\Sigma\ P\&L}{n_{trades}},\quad \text{profit factor} = \frac{\Sigma\ \text{wins}}{|\Sigma\ \text{losses}|}$$

Segmented by strategy family, underlying, and DTE band. **The current month is excluded from every monthly statistic** — a partial month flatters or damns the record at random.

10.5 A stated gap [S]

The system gates statistics on a **minimum n** rather than reporting a **confidence interval**. A binomial proportion interval (Wilson [29] is the standard choice for small n, and is materially better-behaved than the normal approximation) would be strictly more informative than a hard gate: "62% ± 18pp on n=21" says more than either "62%" or nothing. This is a known, deliberate simplification and a documented candidate for future work — it is recorded here rather than quietly omitted.

10.6 Income statistics: what a monthly series can and cannot support [S]

Per-trade statistics (§10.4) do not answer *"can I live off this?"*, because that question is about **time and capital**, not about trades. The monthly series carries its own measures.

Consistency, for a negatively-skewed series. The average is the wrong headline. The decisive quantity is how much of the good months one bad month removes:

$$m_{erased} = \frac{|\min_i Pi_i|}{Pi^+},\quad Pi^+ = \frac{1}{|\{i : Pi_i > 0\}|}\Sigma_{Pi_i>0} Pi_i$$

— the worst month divided by the average *winning* month. On a short-vol book this ratio, not the mean, decides whether the income is spendable.

Return on deployed margin, not on equity. $Pi_{month} / \Sigma_i IM_i$ where

IM_i is each closed trade's initial margin **at entry** (§8.3, evaluated at the index recorded at entry). Trades predating entry-context capture cannot have this reconstructed, so they are **excluded and the coverage percentage is reported** rather than imputed. The denominator is deliberately margin turned over, not account equity, which this data cannot know.

Drawdown on the realized curve. Cumulative realized P&L over complete months, peak-to-trough, plus whether it recovered and in how many months. Stated explicitly as a **realized** curve: open positions are not marked, so a smooth line here is compatible with a large unrealized loss not yet taken — the exact failure mode a "monthly income" framing invites.

The gate. Monthly statistics need **months**. A hundred trades inside three months is three data points; below a minimum of *complete* months every derived statistic is withheld and the reason printed. The in-progress month is excluded from all of them and shown separately.

Pace, as a multiple rather than a difference. With gap G and d days remaining, against a normal daily rate $Pi_{month}/30$:

$$pace\ multiple = (G/d)/(Pi_{month}/30)$$

A multiple is a statement about *pressure* and a difference is not. Thresholds at 1.5× and 2.5× separate "decide now that you will accept missing it" from "this is how size creeps up and strikes drift toward spot". The check runs **after** two prior gates — is the target itself within W_{pres} (§10.2a), and is there margin headroom to add at all — because if either fails, pace is irrelevant.

11. Risk sizing and backtest method

11.1 Volatility targeting [L] — ref. [30]

$$w = min(1, (\sigma_{target})/(\sigma_{realized})),\ w \geq w_{min}$$

Exposure shrinks when the market gets loud. Moreira & Muir [30] is the reference result that volatility-managed exposure improves risk-adjusted returns; the floor w_{min} is this system's own addition, to stop the rule from sizing to zero in a spike. **[S]**

11.2 Walk-forward backtesting, no lookahead [S]

The invariant, enforced by a strict slicing convention: the "known world" at day t is the slice **up to and including** t; forward outcomes are measured on **strictly later** bars. IV-rank percentiles are computed only over the trailing window of *past* data. Any accidental use of a future bar inflates results, so the slice boundaries are what the tests guard — verified against **seeded ground truth** so the absence of lookahead is provable rather than asserted.

11.3 Empirical findings from this book's own data [M]

- The **ATM** vol-risk premium is razor-thin (implied ≈ realized-forward over ~30 days).

The edge is in the OTM skew/tail, not ATM richness.

- **IV-rank timing does add** roughly a couple of vol points — "sell when IV rank is high"

is validated. The middle tier is noisy from overlapping windows (§5.2) and is reported as such rather than smoothed.

- Crypto **tails run ≈2.7× Gaussian** at monthly horizons: a 2σ/30-day put finishes

in-the-money ~6% of the time against ~2% under normal theory. Consistent with the long-standing stylized facts on heavy-tailed returns (Mandelbrot [31]; Cont [32]), and a direct argument against long-dated naked puts at wide-but-not-wide-enough strikes.

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Exchange documentation [X]

The fee, margin, delivery, contract-size, and settlement mechanics in §8 are taken from the venues' own published help pages and instrument metadata, and are **re-verified against the live instrument list** by an automated check rather than trusted from a constant in the code. Where a parameter could not be confirmed from a primary source, it is carried as an explicitly flagged placeholder and the affected output emits a warning.

On the practitioner constructs [P]

§7 (max pain, GEX, zero-gamma flip) has no canonical academic definition. The system uses these constructs because the flow mechanism behind them is real and documented ([17]–[21]), but treats their *predictions* as hypotheses under test rather than inputs — which is the entire purpose of the pin-history store (§7.4).

Disclaimer

Nothing in this document is financial advice. Formulas are reproduced for transparency about what the system computes; their presence does not imply that any strategy derived from them is profitable. Short-premium options trading carries the risk of large and, in some structures, unbounded losses.